

DelegationChain: Parameter-Bound Delegation Chains for Cross-Organizational Agent Authorization

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Abstract

Autonomous agents built on large language models increasingly invoke tools across organizational boundaries, frequently via the Model Context Protocol (MCP). The receiving organization has no verifiable guarantee that such an invocation was authorized, that its specific parameter values were sanctioned rather than just its action type, or that the delegation chain behind it is intact. We present DelegationChain, a protocol for parameter-bound delegation chains between mutually distrusting organizations. Each hop is signed over a recursive digest of the chain, and the invocation binds exact parameter values. A typed policy language, with closed-world parameter declarations and mandatory audience clauses, admits a polynomial-time containment check that is sound with respect to its semantic subset relation. A federated registry provides proof-of-possession registration, policy ceilings pinned by the receiving organization, and revocation by identity–key binding. We implemented the protocol in Rust and evaluated it with a threat-mapped security suite, a differential test of the containment procedure (no soundness violations in 108,370 well-formed scope pairs), and a benchmark whose analysis plan was frozen before measurement. The benchmark also tests the design choice the protocol was first built around: aggregating BLS signatures across the chain. Aggregation is not a net benefit in any configuration we measured. With prefix caching, a BLS verifier is 12.2–24.2 $\times$ slower than an Ed25519 one, and when every call carries a new chain 4.9–16.0 $\times$ slower, while aggregation saves at most 8.8% of a chain’s bytes, because the token bodies, with their keys and policies, dominate it. We therefore specify DelegationChain over per-hop Ed25519 signatures, retaining aggregation as a measured variant; on our hardware this instantiation verifies a three-agent chain under the paper’s example policy in 109 μ s, or 26 μ s when the delegation prefix is cached.

1 Introduction

Software agents built on large language models are increasingly deployed in configurations where they invoke external tools and delegate sub-tasks to other agents. The Model Context Protocol (Model Context Protocol Project, 2026), introduced in late 2024, has emerged as a widely-adopted standard for this interaction pattern. As deployment scales, an architectural pattern is becoming common: agents operated by one organization invoke tools or delegate work to agents operated by another.

Cross-organizational agent invocation creates an authorization problem that existing infrastructure does not adequately address. Consider an agent at Organization A, acting on behalf of a

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human principal, that needs to invoke a payment-processing tool at Organization B. Organization B’s tool must decide whether to honor the request. To do so safely, B needs verifiable answers to four questions: *(i)* did the request actually originate from an authorized agent at A, rather than from an attacker impersonating that agent; *(ii)* was the agent operating within the scope its human principals authorized, rather than acting beyond its delegated authority; *(iii)* were the specific parameter values supplied — the amount, the recipient, the file path — sanctioned by the authorization chain, or has an intermediate party substituted them; and *(iv)* has the authorization chain itself been preserved intact, or has some step in the chain been forged, dropped, or reordered.

Existing protocols address subsets of these questions but do not jointly satisfy them. OAuth 2.1 (Hardt et al., 2026) proves an agent’s identity to a resource server but does not bind authorization to specific parameter values, and where tokens are introspected rather than self-contained it requires a synchronous call to the authorization server on the verification path. Its answer to crossing organizational boundaries, the identity-chaining draft (Schwenkschuster et al., 2026), narrows scope at each boundary but costs a round trip to the authorization servers on both sides of every one. Macaroons (Birgisson et al., 2014) provide attenuated capability delegation with policy caveats and reach across trust domains through third-party caveats, at the cost of a round trip to each third party to collect a discharge. SPIFFE (SPIFFE Project, 2026) provides workload identity within a trust domain but does not include parameter binding or delegation chains. UCAN (UCAN Working Group, 2026) extends capability delegation to public-key settings and, in version 1.0, specifies a policy language over command arguments, but it defines no containment relation between capabilities and does not address MCP integration. Among MCP-specific work, AgentBound (Bühler et al., 2026) provides single-organization runtime enforcement, measurement work (H. Zhou et al., 2026) establishes the size of the gap, and Maloyan and Namiot (2026) propose a protocol extension that authenticates servers and their messages to clients; none provides a cross-organizational proof that an agent’s invocation was authorized. The Agent Identity Protocol (Prakash, 2026a) is the closest design: it offers locally verifiable delegation chains built on Biscuit, but its tokens do not bind an invocation’s parameter values, and it has no registration or federation model and defers revocation. ACLE-MCP (Ding et al., 2026) issues a short-lived lease for each protected call that carries parameter bounds and attests the provider-side workload; its leases are issued per call rather than verified from a chain built once per delegation.

This paper presents DelegationChain, an authorization protocol for cross-organizational agent invocation. A receiving organization should be able to verify an invocation on its own: that it is bound to the exact parameter values of the call, that it was checked against policies both organizations can reason about, and that the delegation chain behind it is intact. DelegationChain provides this with three components. Delegation chains are built on a recursive chain digest and signed hop by hop, so that each signature commits to every earlier hop. A typed policy language, with closed-world parameter declarations and mandatory audience clauses, has a containment check that is polynomial and sound with respect to its semantic subset relation. A federated registry provides proof-of-possession registration, policy ceilings pinned by the receiving organization, and revocation by identity–key binding.

We implemented the protocol, tested it against its threat model and against a differential oracle for its policy language, and measured it under an analysis plan fixed before any measurement. The measurement also settled the design choice the protocol was first built around: aggregating BLS signatures (Boneh, Lynn, et al., 2001; Boneh, Gentry, et al., 2003) across the chain, so that a chain of any length carries one 96-byte signature. Aggregation is not a net benefit in any configuration we measured. With prefix caching a BLS verifier is 12.2–24.2× slower than an Ed25519 one, and aggregation saves at most 8.8% of a chain’s bytes. We therefore specify DelegationChain over per-hop Ed25519 signatures (Josefsson and Liusvaara, 2017), and keep

aggregation as a measured variant.

Contributions. This paper makes five contributions.

1. *A threat model* for cross-organizational agent invocation, organized into six families of attack and separating what the protocol prevents from what it only bounds (Section 3).
2. *A protocol* for parameter-bound delegation chains, built from a recursive chain digest and per-hop signatures, with closed-world parameter declarations and mandatory audience clauses, policy ceilings pinned by the receiving organization, and revocation by identity-key binding. It is specified to the level of a reference implementation (Sections 4 and 5).
3. *A policy language*, typed, whose containment procedure is polynomial and sound with respect to its semantic subset relation, validated by a differential oracle (Sections 6 and 8.2).
4. *A reference implementation and its security evaluation*: a threat-mapped security suite, concurrent-replay tests, cache-equivalence tests and fuzzing (Sections 8.1 and 8.2).
5. *A performance evaluation under a frozen plan*, including a negative result: BLS aggregation does not pay for agent delegation chains. We explain why, and give the cost of the recommended instantiation (Sections 8.3 to 8.5).

We report the aggregation result in full because the design is a natural one: compact signatures over multi-hop chains are an obvious goal, and the cost of reaching it is not visible until the alternatives are measured side by side.

Scope and limitations. The protocol is implemented and evaluated, but not formally verified. The security analysis consists of proof sketches; symbolic verification in Tamarin or ProVerif remains future work (Section 9.3). The performance results come from one machine, and are reported with the frozen plan and every deviation from it (Section 8.3).

Paper organization. Section 2 reviews MCP, the two signature schemes, and capability-based authorization. Section 3 defines the threat model. Section 4 describes the protocol: chain digest construction, parameter binding, approval receipts, the verifier algorithm, encoding, and the aggregate variant. Section 5 describes the federated identity infrastructure, and Section 6 the policy expression language. Section 7 sketches the security argument. Section 8 presents the implementation and its evaluation. Section 9 discusses related work, limitations and future work, and Section 10 concludes. Section A lists the security suite.

2 Background

This section reviews the cryptographic and protocol background required to follow the design. Readers familiar with MCP, Ed25519 and BLS signatures may skim.

2.1 Model Context Protocol

The Model Context Protocol (MCP) (Model Context Protocol Project, 2026) is a JSON-RPC-based protocol standardizing how AI agents discover and invoke external tools. An MCP *host* (typically an LLM-based agent) communicates with one or more MCP *servers*, each exposing a set of tools. Tools are described by name, a typed parameter schema, and documentation. The

host requests a tool invocation by sending a JSON-RPC message specifying the tool name and parameter values; the server executes the tool and returns the result.

MCP’s design assumes that the host and server are mutually trusted, an assumption appropriate for the single-organization deployments common in its early adoption. As MCP deployments scale to cross-organizational configurations, this assumption becomes untenable. A server invoked by an unknown agent has no cryptographic basis on which to decide whether the invocation is authorized. The protocol’s current security guidance relies on transport-layer authentication (typically OAuth) which addresses caller identity but not the broader authorization questions enumerated in Section 1. The gap is not hypothetical: a measurement study of real-world remote MCP servers (H. Zhou et al., 2026) identified 7,973 live servers and found that 40.55% expose tools without any authentication at all, and that among the OAuth-enabled servers it was able to probe, every one exhibited at least one authentication flaw.

That measurement predates the current revision of the specification, 2026-07-28 (Model Context Protocol Project, 2026). The revision deprecates Dynamic Client Registration, the source of the most prevalent flaw class Zhou et al. report, in favor of Client ID Metadata Documents. It also makes the core protocol stateless, removing the initialization handshake so that every request carries its own protocol version and client information. Neither change touches the question this paper addresses. The client information a request carries is self-declared, and authorization remains a matter of OAuth tokens scoped to a server, not of evidence that a particular call was sanctioned.

2.2 Signature Schemes

Ed25519. Ed25519 is the instance of EdDSA over the twisted Edwards form of Curve25519, specified in RFC 8032 (Josefsson and Liusvaara, 2017). Public keys are 32 bytes, signatures 64 bytes, and verifying a signature is one signature check, with no pairing. Verification comes in variants that differ in which encodings they accept. RFC 8032 requires the scalar S of a signature to be reduced ($S < L$), and with that check Brendel et al. (2021) prove Ed25519 strongly unforgeable (SUF-CMA). Rejecting small-order points as well, in the public key and in the signature’s R , adds resistance to key-substitution attacks. The default instantiation of DelegationChain signs each hop with Ed25519 and verifies with strict verification, which performs these checks (Section 4.2).

BLS and aggregation. The Boneh–Lynn–Shacham (BLS) scheme (Boneh, Lynn, et al., 2001) is the scheme behind DelegationChain’s aggregate variant (Section 4.8). It is built from bilinear pairings on elliptic curves, and is specified for interoperable use by Boneh, Bradley, et al. (2026) on the BLS12-381 curve (Sakemi et al., 2026). A signature is a single point in $\mathbb{G}_2$ (96 bytes compressed) and a public key a point in $\mathbb{G}_1$ (48 bytes compressed); verifying one signature is a pairing check.

Its attraction is *aggregation*. Given signatures $\sigma_1, \dots, \sigma_n$ on distinct messages $m_1, \dots, m_n$, their sum $\sigma_{\text{agg}} = \sum_i \sigma_i$ is itself a point of $\mathbb{G}_2$, and its validity with respect to the public keys and messages is checked by a single multi-pairing. Aggregate signatures of this kind were introduced, and proved unforgeable for distinct messages, by Boneh, Gentry, et al. (2003); the CFRG draft’s basic scheme adopts the same rule, rejecting any aggregate in which two messages coincide (Boneh, Bradley, et al., 2026). The precondition is that the *messages* be distinct, not the signers, which matters for delegation chains, where the same agent may legitimately appear at more than one hop. The recursive chain digest of Section 4.3 guarantees distinct messages at every hop by construction, since each m_k commits to m_{k-1} .

Aggregation replaces n signatures with one 96-byte $\mathbb{G}_2$ element. It also makes verification

cheaper than verifying n BLS signatures independently, because the Miller loops of a multi-pairing share part of their work and need a single final exponentiation. It does not make it $O(1)$: with distinct messages the cost still grows with n . The better-known hazard of BLS aggregation, the rogue-key attack, arises when signers sign a *common* message. Defenses for that setting include proof-of-possession at registration (Ristenpart and Yilek, 2007) and the alternative analyzed by Boneh, Drijvers, et al. (2018); neither is needed here, because distinct messages are themselves the rogue-key defense of the basic scheme (Boneh, Bradley, et al., 2026).

Whether aggregation’s savings outweigh the cost of BLS verification is an empirical question; Section 8 answers it.

2.3 Capability-Based Authorization

The protocol design draws on the capability-based security tradition (Miller, 2006). In a capability system, authority is represented by unforgeable tokens (capabilities) that the holder can present to obtain access. Capabilities can be *attenuated*: a holder may produce a restricted version of a capability and delegate the restricted version to another party, without being able to grant authority exceeding what they themselves hold.

Macaroons (Birgisson et al., 2014) apply this model to web authorization. A macaroon is an HMAC chain in which each link adds “caveats” — predicates restricting the macaroon’s use. First-party caveats are checked by the target service itself, and verification recomputes the HMAC chain using a secret that the target service knows.

Macaroons do address the cross-domain case, through *third-party caveats*. A caveat may name a principal other than the target service; to use the macaroon, the holder must obtain a *discharge macaroon* from that principal attesting that the caveat is satisfied, and present it alongside. The discharge is cryptographically bound to the macaroon it discharges, and the target service verifies it without sharing a secret with the third party. It is therefore not correct to say that Macaroons cannot cross organizational boundaries, and we do not say so.

The difference that matters for our setting is where the work happens. A third-party caveat costs a round trip: before presenting the macaroon, the holder must contact each named third party and collect a discharge, and the third party must be reachable at that moment. DelegationChain places the equivalent authority in signatures the verifier checks directly against registry-resolved public keys, with no discharge collection step on the request path.

UCAN (UCAN Working Group, 2026) extends capability attenuation to public-key settings, signing DAG-CBOR-encoded tokens in a Varsig-headed envelope with Ed25519, P-256 or secp256k1. UCAN is designed for Web3 contexts and lacks features specific to our setting: a containment relation between capabilities (UCAN 1.0 does specify a policy language over command arguments), and MCP transport binding.

DelegationChain inherits the conceptual framework of attenuated capabilities from this tradition. Our contribution is the combination of public-key verification (extending Macaroons’ attenuation model across organizational boundaries), parameter binding to the specific invocation, an explicit policy language with termination and containment guarantees, and a federated identity infrastructure.

3 Threat Model

This section defines the threat model under which DelegationChain operates. We state the protocol’s assets, the assumed adversary capabilities, the threats we defend against, the threats we explicitly do not defend against, and the security properties we seek to provide.

3.1 Assets and Principals

The assets we seek to protect are: (i) the integrity of tool invocations, in the sense that an accepted invocation faithfully represents the authorization granted by the originating principals; (ii) the provenance of authorization chains, in the sense that the chain delivered to a verifier accurately records the parties that authorized each step; and (iii) the integrity of audit records derived from accepted chains.

The principals participating in the protocol are: the *human operator*, who initiates a request and whose authority is the ultimate source of permissions; the *orchestrator agent*, which receives the initial authorization and may delegate to other agents; *specialist agents*, which receive delegated authority and either invoke tools directly or delegate further; the *tool* or *MCP server*, which receives authorization chains and decides whether to honor invocations; the *signing service*, a hosted component that holds an agent’s private key and applies policy enforcement before signing (it receives the body it is asked to sign, with the predecessor digest m_{k-1} , and computes m_k itself, since a service shown only a digest could enforce nothing); the *approval service*, which obtains human approval for actions requiring it and signs approval receipts; the *issuance service*, which signs session tokens for its organization; and the *registry*, which issues identity certificates binding each of these principals to a public key and a role. At minimum, a signing service refuses to sign a body that does not name its own agent’s identifier and key as the signer.

3.2 Adversary Capabilities

We model a probabilistic polynomial-time adversary with the following capabilities. The adversary can observe and modify all network traffic between principals (a standard Dolev-Yao adversary on the network). The adversary can compromise an agent’s runtime environment, gaining the ability to instruct the signing service to sign messages of the adversary’s choice within the scope permitted by policy. The adversary can inject content into agent inputs to attempt prompt injection attacks. The adversary cannot, by default, compromise hardware security modules, the signing service infrastructure, the approval service infrastructure, the registry, or human approvers’ devices; threats involving these compromises are addressed separately and explicitly.

3.3 Threats Within Scope

The threats in scope fall into six families. Variants within a family share a *target* — what the attacker is trying to subvert — and not necessarily a defense; Section 7.3 maps each variant individually to the theorem or mechanism that answers it. Threats are labelled for cross-reference in the security analysis (Section 7).

We divide them by what the protocol actually does about them rather than by family. Section 3.3.1 lists what it *prevents*. Section 3.3.2 lists what it only *bounds*: the attack remains possible, and the protocol limits what it can achieve. Both are in scope, in the sense that the design is answerable for them, but a reader who took the second group for the first would credit the protocol with guarantees it does not provide. The division cuts across two families — T4a and T5b are bounded while the rest of T4 and T5 are prevented — and we would rather have an untidy grouping than a tidy overclaim.

3.3.1 Threats the protocol prevents

T1 — Forgery of tokens and chains. **T1a: *Token forgery*.** An adversary attempts to produce a valid authorization token without holding the corresponding signing key.

T1b: *Chain forgery and reordering.* An adversary attempts to construct an authorization chain that was not produced by the legitimate delegation sequence, including reordering hops, dropping hops, or inserting unauthorized hops.

T2 — Misrepresentation of granted authority. The two variants share a target — the authorization that reaches the verifier is not the authorization that was granted — but not a defense: T2a is defeated by the signature over the parameter encoding, T2b by the containment check.

T2a: *Parameter substitution.* An adversary, positioned between a signer and a verifier, attempts to alter parameter values in an authorized invocation (e.g., changing the destination of a transfer) while preserving the appearance of valid authorization.

T2b: *Scope escalation.* An agent attempts to delegate more authority than it was itself granted.

T3 — Reuse of valid tokens. **T3a:** *Replay across sessions.* An adversary captures a valid authorization chain and attempts to use it after its intended validity window has expired.

T3b: *Replay within validity window.* An adversary captures a valid authorization chain and attempts to use it more than once within its validity window.

T3c: *Cross-context token reuse.* An adversary attempts to reuse a token issued in one context (e.g., for a delegation) as if it had been issued in another context (e.g., for an invocation).

T3d: *Cross-verifier reuse.* An adversary presents a chain that one verifier accepted to a different verifier that accepts the same partner policy and exposes the same tool and action names. Examples are a production and a staging deployment, two regions with separate nonce caches, or two organizations that both pinned the same published policy.

T4 — Subversion of the approval path. The remaining variant of this family, T4a, is bounded rather than prevented and appears in Section 3.3.2.

T4b: *Approval routing attack.* An agent routes approval requests to a permissive approval service when policy requires a stricter one.

T4c: *Approval reuse across invocations.* An adversary attempts to reuse a valid approval receipt with an invocation different from the one approved.

T4d: *Stale approval after parameter modification.* An adversary attempts to modify invocation parameters after approval has been obtained.

T5 — Attacks on the identity infrastructure. The variant T5b is bounded rather than prevented and appears in Section 3.3.2.

T5a: *Key misattribution at registration.* An adversary attempts to have the registry certify, under the adversary’s own identifier, a public key it does not control — typically another principal’s. The adversary gains no ability to sign. What it breaks is the binding between keys and parties that non-repudiation depends on: the key’s real holder can disown signatures made under it.

T5c: *Revocation suppression.* An adversary suppresses revocation information from reaching verifiers, extending the window during which compromised credentials remain accepted.

T5d: *Certificate substitution.* An adversary substitutes a different certificate for the same identity during the registration or resolution process.

T5e: *Role confusion.* An adversary holding a key certified for one role — an agent, a signing service or an approval service — uses it to sign in another. For instance, it signs a session token naming itself as issuer, to obtain its organization’s full policy ceiling rather than the authority it was granted.

3.3.2 Threats the protocol bounds but does not prevent

The protocol limits the damage the following attacks can do, but does not prevent them, and we do not claim that it does. Section 7.3 states what the bound is in each case.

T4 — Subversion of the approval path (one variant). **T4a: *Compromised approval service.*** An adversary obtains the signing key of an approval service and issues fraudulent approvals. The protocol cannot prevent this: a valid signature under a registered approval key is, by construction, a valid receipt. What it does is contain the consequences. Approval receipts are verified separately from the chain signatures (Section 4.5), and an approver’s certificate is accepted only for receipts, never for a session or delegation body (Section 4.6). A compromised approval service therefore cannot forge delegation chains, cannot enlarge any agent’s scope, and cannot attest to anything beyond what it was already trusted to attest to; the blast radius is exactly the set of invocations that policy routes through that service. Detection and recovery belong to the identity infrastructure — transparency log monitoring, revocation and rotation (Section 5). Policies name approval services by identifier, and their keys are resolved and checked for revocation at verification time, so revoking a compromised approval key takes effect without a new policy version.

T5 — Attacks on the identity infrastructure (one variant). **T5b: *Compromised registry signing key.*** An adversary obtains a registry’s root signing key and issues fraudulent certificates. The protocol cannot prevent this, and the security arguments of Section 7 assume an honest root (Section 7.1). The damage is bounded in two ways. A registry root is trusted only for identifiers in its own organization’s namespace (Section 5.4), so the compromise cannot reach another organization’s identities. And sessions obtained through certificates it mints stay within the policies verifiers have pinned for that organization (Section 6). Recovery requires re-establishing the root out of band (Section 5.5).

T6 — Misuse of an agent within its own policy. **T6a: *Prompt injection within policy.*** An adversary, controlling content fed to an agent, induces the agent to take actions that are within the agent’s permitted policy but contrary to the principal’s intent.

T6b: *Confused deputy in delegation.* An agent is induced to sign a delegation token favoring an adversary by manipulation of the inputs the agent uses to construct delegation decisions.

3.4 Threats Outside Scope

The following threats are explicitly outside our scope. Defending against them requires mechanisms orthogonal to the protocol.

A fully compromised signing service can produce signatures on any messages within an agent’s policy; the protocol cannot prevent this and relies on operational hardening (HSM-based key custody, monitored access). Compromise of an approval service is treated in Section 3.3.2 rather than here, because the protocol does bound its effect even though it cannot prevent it. Side-channel attacks on cryptographic implementations are out of scope; we rely on the security properties of well-audited signature libraries such as `ed25519-dalek` and `blst`. Attacks on the

human approver’s device or authentication mechanism are out of scope; the protocol assumes a trusted path between the approval service and the human. Denial-of-service attacks on network infrastructure are out of scope. Post-quantum security is out of scope for this version; neither Ed25519 nor BLS12-381 is post-quantum secure, and migration to a post-quantum scheme is left to future work.

3.5 Security Properties Sought

We seek to provide the following security properties, each of which is formally stated and argued in Section 7:

Authenticity. Authorization tokens are unforgeable: no PPT adversary lacking access to a signing key can produce a token that verifies under the corresponding public key, except with negligible probability.

Parameter binding. A token authorizing an invocation with parameter values P is invalid for any parameter values $P' \neq P$.

Chain integrity. The signatures over a delegation chain cover the entire chain content; modification, reordering, truncation, or extension of the chain invalidates them.

Scope monotonicity. In any valid chain, the session scope is contained in a policy the verifier has pinned for the issuing organization, and the scope at hop $k + 1$ is contained in the scope at hop k .

Non-repudiation. Neither party to a verified action can plausibly deny it, assuming honest signing key management.

Bounded revocation propagation. Once a registry has issued a revocation assertion, the additional window before verifiers cease accepting tokens under the revoked binding is bounded by a configurable parameter (default: 120 seconds). Detection of the compromise and its reporting to the registry precede this window and are properties of the deploying organization’s monitoring rather than of the protocol. The protocol bounds propagation, not total exposure.

4 Protocol Design

This section presents the cryptographic core of DelegationChain: the participants, the token types, the chain digest construction that binds each hop’s signature to every earlier hop, the encoding of parameter-bound authorization, the handling of approval receipts, the verifier algorithm, and the aggregate variant. We describe the design at a level of detail sufficient to support the security analysis of Section 7 and the reference implementation of Section 8. Bit-level details beyond those given here — field numbering within each structure, and the exact byte layout of the wire format — are not fixed by this paper; the specification published with the reference implementation fixes them for that implementation (see the data and code availability statement).

4.1 Participants and Tokens

A DelegationChain authorization chain has the following participants, in the order they contribute to the chain. The *issuer* I is the organization’s issuance service, holding a certificate of kind **issuer** from its registry, which issues an initial session token to the orchestrator agent at the start of an authorized session. The registry’s root key certifies the issuer but never signs sessions

itself, so it can stay offline. The *orchestrator* A_1 is the agent that receives the session token and initiates the workflow. Intermediate *specialist agents* $A_2, \dots, A_{N-1}$ may be delegated sub-tasks. The *invoking agent* A_N is the agent that finally invokes a tool at the verifier. The *verifier* V is the receiving tool or MCP server. When policy requires human approval, an *approval service* contributes a separately-signed receipt.

The chain contains three token types. The *session token*, signed by the issuer, authorizes the orchestrator to operate within a scope for a bounded duration. *Delegation tokens*, signed by intermediate agents A_k for $1 \leq k < N$, attenuate authority and pass it to the next agent. The *invocation token*, signed by the final agent A_N , binds the invocation to specific tool, action, and parameter values. The chain delivered to the verifier consists of all token bodies and one signature per hop; under the aggregate variant (Section 4.8), a single aggregate signature replaces the per-hop signatures.

4.2 Cryptographic Primitives

DelegationChain is parameterized by a signature scheme $\Sigma = (\text{KeyGen}, \text{Sign}, \text{Verify})$, assumed existentially unforgeable under chosen-message attack. Every signed message is a 32-byte SHA-256 digest. We specify two instantiations. The default signs each hop with Ed25519 (Josefsson and Liusvaara, 2017) and verifies with strict verification, which rejects non-canonical encodings and small-order points (Section 2.2). The aggregate variant (Section 4.8) uses BLS signatures on BLS12-381 and aggregates them across the chain. Within one deployment, registries, approval services and chain signers use the same scheme, so certificates and receipts are signed with Σ too.

SHA-256 is used for all non-signature hashing, including the chain digest construction.

4.3 Chain Digest Construction

The central cryptographic construction is a recursive digest chain binding each hop to all prior hops. We use domain separation tags to ensure tokens of different types cannot be confused (preventing cross-context reuse, T3c). Tags are 8-byte ASCII strings with a null terminator: TAG_SES for sessions, TAG_DEL for delegations, TAG_INV for invocations.

Let $\text{Canon}(\cdot)$ denote canonical encoding (deterministic CBOR per RFC 8949 §4.2.1, with the additional rules summarized in Section 4.7).

Session message. The issuer signs:

$$m_0 = H(\text{TAG_SES} \parallel \text{Canon}(\text{SessionBody})) \quad (1)$$

where SessionBody contains the issuer’s identifier and public key (`issuer_id`, `issuer_pk`), the orchestrator’s identifier and public key (`subject_id`, `subject_pk`), a session identifier (`session_id`), a hash committing to the governing policy (`policy_hash`), a scope expression, issue and expiry timestamps, and a fresh nonce. The issue timestamp is informational. No check in the verifier depends on it, since, like every timestamp a signer supplies, it is not evidence of when signing happened (Section 5.6).

Delegation messages. For each intermediate hop A_k with $1 \leq k < N$:

$$m_k = H(\text{TAG_DEL} \parallel m_{k-1} \parallel \text{Canon}(\text{DelegationBody}_k)) \quad (2)$$

where DelegationBody _{k} contains the delegator’s identifier and public key (`delegator_id`, `delegator_pk`), the delegatee’s identifier and public key (`delegatee_id`, `delegatee_pk`), the attenuated sub-scope, its hop index k (`hop_index`), the parent session’s identifier (`session_id`), expiry, and a

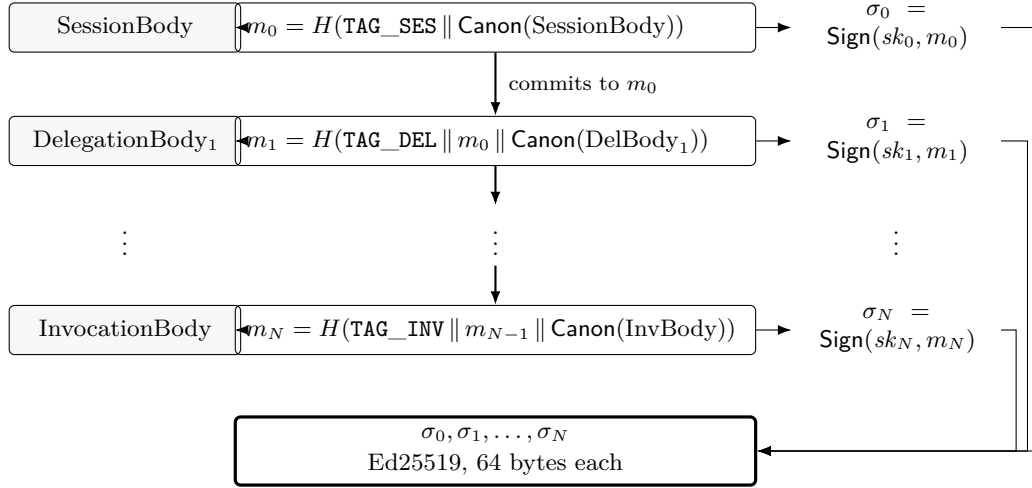


Figure 1: The recursive chain digest. Each hop’s message commits to its predecessor’s, so the order of hops is fixed by the digests themselves; the domain separation tag fixes each token’s kind. Each hop signs its own message, and the chain carries every signature. The verifier recomputes $m_0, \dots, m_N$ from the received bodies, using the tag determined by each token’s position, and checks each signature against its message. Under the aggregate variant (Section 4.8) the signatures are summed into one 96-byte $\mathbb{G}_2$ element.

fresh nonce. The hop index and session identifier are redundant with the digest recursion, which already fixes each delegation’s position and session. The verifier checks them anyway, in phase 1, because doing so rejects a chain spliced from two sessions, or with a hop out of place, before any signature is verified. The delegator’s identifier is carried explicitly so that a verifier can resolve the signer’s certificate from the body it is checking, rather than having to reach back into the preceding token. The critical property is that m_k commits to m_{k-1} via the hash, recursively committing to the entire chain history.

Invocation message. The final hop A_N signs:

$$m_N = H(\text{TAG_INV} \parallel m_{N-1} \parallel \text{Canon}(\text{InvocationBody})) \quad (3)$$

where `InvocationBody` contains the invoker’s identifier and public key (`invoker_id`, `invoker_pk`), the identifier of the verifier the invocation is addressed to (`aud`, a `service` identifier; Section 5.2), the target tool and action identifiers, the canonically encoded parameter values, a hash of the parameters (defense-in-depth against encoding bugs), one approval receipt for each approval service the policy requires (possibly none; Section 4.5), and timing and nonce fields.

Figure 1 shows the construction as a whole.

Signatures. Each participant signs its message: $\sigma_k = \text{Sign}(sk_k, m_k)$, for $k = 0, \dots, N$, where sk_0 denotes the issuer’s key sk_I , since the issuer signs the session body. In the default instantiation the chain carries $\sigma_0, \dots, \sigma_N$, one 64-byte Ed25519 signature per hop. Because m_k commits to m_{k-1} , each σ_k authenticates the entire chain prefix up to hop k ; a verifier that has already verified a prefix can therefore cache that result and, for a later chain with the same prefix, verify only the new hop’s signature (Section 8.4).

Role distinguishability by position. The verifier reconstructs the m_k values from received token bodies using the tag determined by token position: `TAG_SES` for the session token, `TAG_DEL` for each intermediate delegation token, and `TAG_INV` for the invocation token. Every body also

carries an explicit `kind` field, with value `session`, `delegation`, or `invocation`, inside its signed encoding. The verifier *rejects* chains where the position-derived tag does not match a body’s `kind` — preventing chain-extension attacks that would attempt to reuse an invocation token as a delegation, or vice versa. Such a body would fail signature verification in any case, since the verifier would hash it under a tag it was not signed with. The explicit field lets the verifier reject it in phase 1, before any signature is verified, and removes any ambiguity about how a body is to be parsed.

4.4 Parameter Binding

Parameter binding is achieved by including the canonical encoding of parameter values directly in the message m_N that the invoking agent signs. Any modification to parameter values — changing an amount, substituting a recipient, altering a file path — changes the canonical encoding, which changes m_N , which invalidates σ_N (or, under the aggregate variant, the aggregate).

To support nested parameter structures (e.g., `order.payment.amount`), parameter values are encoded as a nested CBOR map with text-string keys sorted lexicographically by encoded form (per RFC 8949 §4.2.1). Parameter values are restricted to unsigned and signed integers, byte strings, NFC-normalized text strings, booleans, null, and nested maps and arrays of the same. Float values and tagged values are disallowed in parameter encodings, to eliminate edge cases around floating-point comparison and the introduction of unexpected types. Parameter map *keys* are text strings and are NFC-normalized on the same terms as text values. Without an explicit rule for keys, two implementations could canonically re-encode the same logical parameter map to different byte strings, and the re-encoding check of Section 4.7 would diverge between them.

Parameter binding is a cryptographic property, not an authorization one. It guarantees that the values a verifier reads are the values the invoking agent signed; it says nothing about whether those values were permitted. Whether they were is decided by the policy evaluation of Section 6.3, whose closed-world treatment of undeclared parameters is what turns binding into constraint.

The `InvocationBody` also includes `params_hash`, a SHA-256 digest of the canonical parameter encoding. The verifier recomputes this hash from the encoded parameters and rejects any chain where it does not match. This is redundant with the signature check but serves as defense-in-depth against subtle encoding bugs in implementations.

4.5 Approval Receipts

When policy requires human approval for an action, the `InvocationBody` carries an *approval receipt* from each approval service the policy requires, signed by that registered service. The receipt binds the human’s approval decision to the specific invocation being approved.

The flow is as follows. The agent constructs the `InvocationBody` *except* for the receipts field, computes the invocation digest of this preliminary body (defined below), and submits the digest along with the full body to the required approval service. The service presents the proposed invocation to a human approver. On approval, the service signs an *approval body* containing the service’s identifier and public key (`approver_id`, `approver_pk`), the invocation digest, opaque attestation data identifying the human approver, and issue and expiry timestamps (`iat`, `exp`). The signed receipt is returned to the agent, which embeds it in the now-complete `InvocationBody` and proceeds with chain construction.

Receipts are signed under their own domain separation tag, distinct from those of chain signatures and registration proofs. The kind check of phase 7 already stops a key from being used across these roles, so the separate tag is defense in depth.

The *invocation digest* is

$$\text{InvocationDigest}(B_N) = H(\text{TAG_IVD} \parallel \text{Canon}(B_N \text{ without its receipts field})).$$

The receipts field is omitted when there are no receipts, so this is exactly the digest of the preliminary body the agent submitted, and the tag separates it from m_N . Receipts are ordered by approver identifier, bitwise ascending, with at most one per approval service. A body that breaks either rule, or carries an empty receipts field, is malformed. A receipt from an approval service that the policy does not require for the invocation is ignored.

The approval signature is verified separately from the chain signatures. This decoupling has two consequences. First, the security argument for chain integrity does not depend on approval service trust: compromise of an approval service does not enable chain forgery. Second, approval verification adds, per receipt, one signature verification (under the aggregate variant’s BLS keys, a hash to $\mathbb{G}_2$ and a two-pairing check). Receipts are typically zero or one per invocation.

Policy can require approvals from multiple services (logical AND); the required service identifiers are part of the policy that the session token’s `policy_hash` commits to. The policy names approval services by identifier only, never by key. Their keys are resolved at verification time like any other signer’s — root, namespace, kind **approver**, expiry and revocation (Section 5) — so revoking or rotating an approval service’s key takes effect without a new policy version and without every partner re-pinning. The policy hash is fixed at session start and does not vary with where the agent sends its approval request, so an agent that routes to a different (perhaps more permissive) service still presents the correct hash. What it cannot present is a receipt from a service the policy requires. The verifier, having resolved the policy that the hash commits to, and having checked that every scope in the chain is contained in it (Section 6), obtains an approval requirement no weaker than the policy’s own, and rejects a chain whose receipts do not cover it. This is the defense against the approval routing attack (T4b).

4.6 Verification

The verifier algorithm proceeds in phases, ordered to fail fast on cheap checks before performing expensive cryptographic operations. Algorithms 1 and 2 state the procedure; the paragraphs that follow comment on individual phases. Every **reject** is terminal. The only decision-relevant state the procedure changes is the nonce cache, in its final step, and that step is an atomic insert-if-absent. Resolution and policy loading may fill caches (Section 5.4). That affects the cost of later verifications but not, within the propagation bound of Section 5.6, their outcome. A chain that fails any check therefore leaves the nonce cache exactly as it found it.

Three abbreviations keep the statement short, all of them reading fields named in Section 4.3. Write $\text{sid}(B_k)$ for the identifier of the party that signed B_k : **issuer_id** for the session body, **delegator_id** for a delegation body, **invoker_id** for the invocation body. Write $\text{spk}(B_k)$ for that party’s public key as named in the body: **issuer_pk**, **delegator_pk** or **invoker_pk**. Every body names its signer’s key, so during a scheduled rotation, when two certificates for one identifier are valid at once, the key determines which one applies. Write $\text{scope}(B_k)$ for the body’s scope expression: the granted scope for the session body, the attenuated sub-scope for a delegation body. Finally, $\text{org}(x)$ is the **organization-id** prefix of an identifier x (Section 5.2); $\text{Root}[o]$ is the registry root key the verifier holds for organization o (Section 5.1); and $\text{Pinned}[o]$ is the set of policy hashes the verifier has accepted from organization o (Section 6). The certificate kind that position k requires is $\text{role}(k)$: **issuer** for $k = 0$, **agent** for $k \geq 1$. *self* is the verifier’s own service identifier.

Structural and temporal checks. The verifier first decodes the received chain and confirms canonical encoding (rejecting non-canonical inputs to prevent encoding-ambiguity attacks), that

Algorithm 1 DelegationChain verification, phases 1–5: structural, temporal, and identity checks. Phases run cheapest-first; every **reject** is terminal. Continued in Algorithm 2.

```

1: procedure VERIFY(received chain  $C$ , current time  $t$ )
  ▷ Phase 1 — structure and encoding
2:    $(B_0, \dots, B_N, \sigma) \leftarrow \text{DECODE}(C)$ ; reject if decoding fails
3:   reject if  $N < 1$ 
4:   for all  $B_k$  do
5:     reject if  $\text{Canon}(B_k) \neq$  received bytes of  $B_k$            ▷ non-canonical encoding
6:   end for
7:   reject if any  $B_k.\text{kind}$  disagrees with the tag its position implies
8:   reject if  $B_N.\text{aud} \neq \text{self}$                                    ▷ addressed to this verifier
9:   reject if  $H(\text{Canon}(B_N.\text{params})) \neq B_N.\text{params\_hash}$ 
10:  for  $k \leftarrow 1$  to  $N - 1$  do
11:    reject unless  $B_k.\text{hop\_index} = k$  and  $B_k.\text{session\_id} = B_0.\text{session\_id}$ 
12:  end for
  ▷ Phase 2 — temporal
13:  reject if  $t \notin [B_N.\text{nbf}, B_N.\text{exp}]$ 
14:  for  $k \leftarrow 1$  to  $N$  do
15:    reject if  $B_k.\text{exp} > B_{k-1}.\text{exp}$            ▷ expiry must not grow along the chain
16:  end for
  ▷ Phase 3 — replay
17:  reject if  $(\text{spk}(B_N), B_N.\text{nonce}) \in \text{NonceCache}$ 
  ▷ Phase 4 — key chain consistency
18:  reject if  $(B_0.\text{subject\_id}, B_0.\text{subject\_pk}) \neq (\text{sid}(B_1), \text{spk}(B_1))$ 
19:  for  $k \leftarrow 1$  to  $N - 1$  do
20:    reject if  $(B_k.\text{delegatee\_id}, B_k.\text{delegatee\_pk}) \neq (\text{sid}(B_{k+1}), \text{spk}(B_{k+1}))$ 
21:  end for
  ▷ Phase 5 — identity resolution
22:  for  $k \leftarrow 0$  to  $N$  do
23:     $\text{cert}_k \leftarrow \text{RESOLVE}(\text{sid}(B_k), \text{spk}(B_k))$ ; reject if unresolvable
24:    reject unless  $\text{cert}_k$  verifies under  $\text{Root}[\text{org}(\text{sid}(B_k))]$ 
25:    reject unless  $\text{cert}_k.\text{registry\_id} = \text{org}(\text{sid}(B_k))$            ▷ registry owns the namespace
26:    reject unless  $\text{cert}_k.\text{kind} = \text{role}(k)$                          ▷ role fixed by position
27:    reject unless  $t \in [\text{cert}_k.\text{nbf}, \text{cert}_k.\text{exp}]$  and  $(\text{sid}(B_k), \text{spk}(B_k))$  is not revoked at  $t$ 
28:     $\text{pk}_k \leftarrow \text{cert}_k.\text{pk}$                                      ▷ equals  $\text{spk}(B_k)$  by resolution
29:  end for

```

the recomputed `params_hash` matches (Section 4.4), that every delegation body carries its own position as `hop_index` and the session’s `session_id`, and that the invocation is addressed to this verifier: the invocation body’s `aud` must equal the verifier’s own `service` identifier. It then checks that the current time is within the invocation’s validity window, that expiry times are monotonically non-increasing along the chain (each hop’s expiry cannot exceed its parent’s), and that the chain has minimum length one.

Replay protection. The verifier maintains a bounded-time nonce cache indexed by (invoker public key, nonce). Chains whose key is already present in the cache are rejected as replays. A chain is accepted only if inserting its key into the cache succeeds, with TTL equal to the chain’s remaining validity window plus a clock-skew tolerance. Line 13 applies no tolerance of its own: a chain is valid exactly within $[B_N.\text{nbf}, B_N.\text{exp}]$ as the verifier’s clock reads it, and senders must allow for skew when they set those bounds. The tolerance in the TTL has a different purpose. Instances of

Algorithm 2 DelegationChain verification, phases 6–8: policy, approvals, and the chain signatures. Continues Algorithm 1; line numbering and the enclosing procedure carry over.

```

    ▷ Phase 6 — policy
30:   reject if  $B_0.\text{policy\_hash} \notin \text{Pinned}[\text{org}(\text{sid}(B_0))]$                                 ▷ verifier's own choice
31:    $S \leftarrow \text{LOADPOLICY}(B_0.\text{policy\_hash})$ ; reject if unavailable
32:   reject if  $\text{CONTAINS}(S, \text{scope}(B_0))$  is false                                          ▷ the ceiling
33:   for  $k \leftarrow 1$  to  $N - 1$  do
34:     reject if  $\text{CONTAINS}(\text{scope}(B_{k-1}), \text{scope}(B_k))$  is false
35:   end for
36:    $d \leftarrow \text{EVALUATE}(\text{scope}(B_{N-1}), B_N.\text{invocation})$ 
37:   reject if  $d = \text{DENY}$ 
    ▷ Phase 7 — approvals
38:   if  $d = \text{ALLOW-WITH-APPROVAL}(svcs)$  then
39:     for all  $s \in svcs$  do
40:        $R \leftarrow$  the receipt in  $B_N$  with  $R.\text{approver\_id} = s$ ; reject if none
41:        $\text{cert}_R \leftarrow \text{RESOLVE}(s, R.\text{approver\_pk})$ ; reject if unresolvable
42:       reject unless  $\text{cert}_R$  passes the phase-5 checks with kind approver
43:       reject unless  $R$  verifies under  $\text{cert}_R.\text{pk}$  over  $\text{INVOCATIONDIGEST}(B_N)$ 
44:       reject if  $t \notin [R.\text{iat}, R.\text{exp}]$ 
45:     end for
46:   end if
    ▷ Phase 8 — chain signatures
47:   recompute  $m_0, \dots, m_N$  from  $B_0, \dots, B_N$  using position-determined tags
48:   reject if  $m_i = m_j$  for some  $i \neq j$                                                 ▷ aggregate variant's precondition
49:   reject unless  $\text{VERIFYCHAIN}(pk_0..pk_N, m_0..m_N, \sigma)$ 
    ▷ Commit
50:   reject unless  $\text{INSERTIFABSENT}(\text{NonceCache}, (pk_N, B_N.\text{nonce}))$                     ▷ atomic
51:   accept
52: end procedure

```

one logical verifier that share the cache may disagree about the time. The added term must be at least that disagreement, so that no instance evicts a chain's entry while another could still accept the chain.

The lookup in phase 3 is only an early filter. What actually prevents replay is that the final insertion is atomic and fails if the key is already present. The lookup and the insertion are separated by identity resolution, which may involve a network call, and by signature verification. Without atomicity, two copies of one chain presented concurrently would both pass phase 3 and both be accepted, on a single instance as readily as across replicas. The nonce cache is kept per verifier. Replay across *different* verifiers is prevented not by the cache but by the audience field: a chain names its verifier in **aud**, and any other verifier rejects it in phase 1. The cache size is bounded by the product of maximum verification throughput and the maximum validity window.

The cache must be a single logical structure. A deployment that places several verifier replicas behind a load balancer, each holding its own independent cache, does not provide replay protection at all: the same chain presented to two replicas is accepted twice, and an adversary who can influence load-balancer routing can replay at will. Replicated verifiers must therefore share a cache or partition it deterministically by invoker public key. This is a coordination dependency on the verification path, and Theorem 5 is stated with it as an explicit hypothesis.

Public key chain consistency. The verifier checks that each body is signed by the party its predecessor named. The session body’s subject identifier and key must equal the signer identifier and key of B_1 , and each delegation body’s delegatee identifier and key must equal those of the body that follows it. Comparing identifiers as well as keys ensures that the party that acts is the one that was named, even where one key is certified under more than one identifier. The signer of a delegation body is its delegator, and that of the invocation body is its invoker, so the rule also covers the shortest chain: when $N = 1$ there are no delegations, and the orchestrator named in the session is itself the invoker.

Scope monotonicity and policy compliance. The verifier first confirms that the session token’s `policy_hash` is one it has pinned for the issuing organization (Section 6). It then obtains the policy document, from its cache or by hash from the issuing registry (Section 5.4), and establishes the containment chain Eq. (6) using the procedure of Section 6.4: the session scope must lie within the policy, and each hop’s sub-scope within its parent’s. Finally it evaluates the invocation against the innermost scope, rejecting if the invocation falls outside it. The pinning check is what gives the first link its force. The issuer signs both the policy hash and the session scope, so without pinning the verifier would be checking the issuer against a policy the issuer had selected.

Approval verification. If policy requires approvals, the verifier finds, for each required approval service, the receipt that names it. It resolves the service’s certificate through the same checks as phase 5, requiring kind `approver`. It then confirms that the receipt verifies under the certified key over the correct invocation digest, and that the current time lies within the receipt’s own validity window. A chain missing any of these is rejected.

Chain signature verification. Finally, the verifier reconstructs $m_0, m_1, \dots, m_N$ from the received bodies (using position-determined tags) and confirms that they are pairwise distinct (line 48). Distinct messages are the aggregate variant’s rogue-key precondition (Section 4.8); the digest recursion makes a repeat impossible short of a SHA-256 collision in both instantiations, so for a well-formed chain the check never fails. It then checks the chain’s signatures, line 49: `VERIFYCHAIN(pk0..pkN, m0..mN,  $\sigma$ )`. For the default instantiation, `VERIFYCHAIN` checks each σ_k against (pk_k, m_k) with strict Ed25519 verification. For the aggregate variant, it is the multi-pairing check of Eq. (5) (Section 4.8). If this check succeeds, all integrity, parameter-binding, and chain provenance properties hold simultaneously.

Cost. The measured costs of both instantiations are in Section 8.4. Under the aggregate variant, the cost is well approximated by $\alpha + \beta N$, with β one hash to $\mathbb{G}_2$ plus one Miller loop; Section 8.4 measures it.

Figure 2 shows why the order matters. The phases are arranged so that the checks a hostile chain is most likely to fail are also the cheapest to run: a malformed or expired chain is rejected on parsing and integer comparison, and only a chain that has survived every structural, temporal, and policy check reaches the chain-signature verification of phase 8. An adversary who wishes to make a verifier do the expensive work must first construct a chain that is well-formed, unexpired, non-replayed, key-consistent, scope-monotone, and policy-compliant — at which point the chain signatures are the only thing left to establish, and the work is warranted.

On a cold verifier there is one exception. Identity resolution (phase 5) verifies each signer’s certificate under its registry root before the policy phase: $N + 1$ signature verifications, each of them a pairing check when registry keys are BLS. Their results are cached per binding, so an adversary naming real identities can force these checks at most once per certified identity per

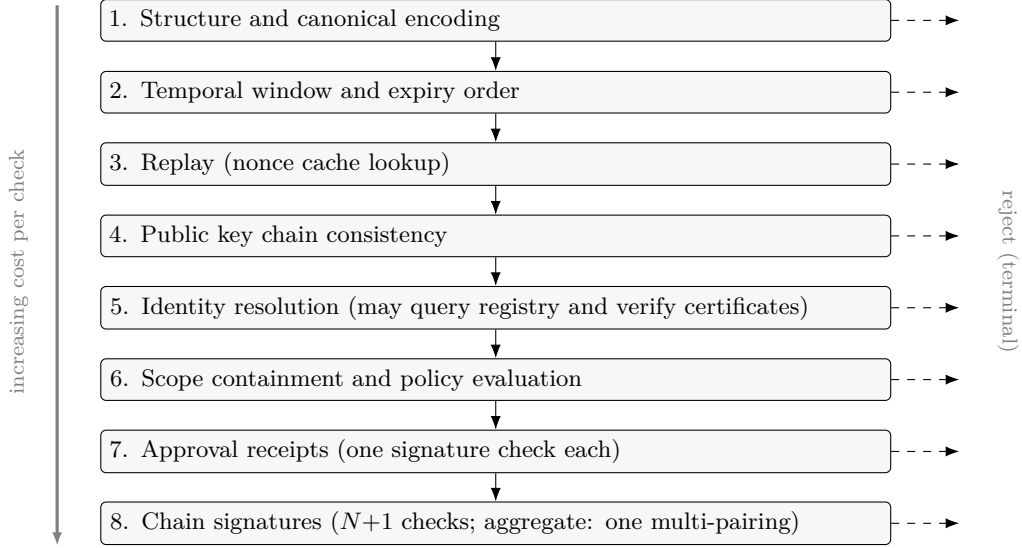


Figure 2: Phase ordering in Algorithms 1 and 2. Cheap checks run first, so that, once certificates are cached, a chain which cannot succeed is discarded before any chain signature is verified; on a cold verifier, phase 5 verifies certificates first. No state that affects a later decision changes until the final phase has passed; certificate and policy caches may be filled earlier.

cache lifetime, and at most $N + 1$ per chain. Failed verifications are not cached, so an adversary on the resolution path who serves invalid certificates can force one on every attempt; deployments should authenticate or rate-limit resolution.

4.7 Encoding

All signed structures use deterministic CBOR encoding per RFC 8949 §4.2.1, with the following additional rules. In *protocol structures* — session, delegation and invocation bodies, certificates, and approval receipts — map keys are integers rather than text strings, eliminating UTF-8 normalization considerations and reducing encoding size. Parameter maps are the exception: they carry names chosen by the tool author rather than by this specification, and so use text-string keys sorted lexicographically by encoded form, per Section 4.4. The scoping is deliberate, and an implementation that applies the integer-key rule to parameter maps will not interoperate with one that does not. Float and tagged values are disallowed in protocol structures except where explicitly required. Ed25519 public keys and signatures use their RFC 8032 encodings (Josefsson and Liusvaara, 2017). A non-canonical signature encoding, an R whose y -coordinate is not reduced or an S with $S \geq L$, is a decoding failure: in a chain or a receipt, line 2 rejects it. A non-canonical or small-order public key is rejected when the structure that carries it as a key is decoded: by the registry at registration (Section 5.3), and by the verifier when it checks a certificate (line 24). Key fields in token bodies are compared as bytes with the resolved certificate’s key and never used as points. Strict verification also rejects a signature whose R is of small order, at line 49. The encoding of BLS group elements is part of the aggregate variant (Section 4.8).

Verifiers reject any structure whose received encoding differs from the canonical re-encoding of the decoded content. This precludes a class of attacks based on adversarially-crafted non-canonical encodings that parse correctly but hash to different values than the implementation expects.

4.8 Aggregate Variant

The aggregate variant replaces the per-hop signatures with a single BLS aggregate. Everything else in the protocol — the digests, the bodies, the verifier’s phases 1–7 — is unchanged; only line 49’s `VERIFYCHAIN` differs.

Parameters. The variant uses BLS12-381 (Sakemi et al., 2026) with the minimal-pubkey-size variant of Boneh, Bradley, et al. (2026): public keys are points in $\mathbb{G}_1$ (48 bytes compressed), signatures are points in $\mathbb{G}_2$ (96 bytes compressed), and the pairing maps $\mathbb{G}_1 \times \mathbb{G}_2$ to $\mathbb{G}_T$. Hash-to-curve uses RFC 9380 (Faz-Hernández et al., 2023) with suite `BLS12381G2_XMD:SHA-256_SSWU_RO_`. With signing key $sk \in \mathbb{Z}_q$ and public key $pk = sk \cdot g_1 \in \mathbb{G}_1$, a message m is signed as $\text{Sign}(sk, m) = sk \cdot \text{HashToG}_2(m) \in \mathbb{G}_2$, and verification checks $e(g_1, \sigma) \stackrel{?}{=} e(pk, \text{HashToG}_2(m))$. Chain signatures use the basic scheme of the CFRG draft (Boneh, Bradley, et al., 2026), whose ciphersuite identifier is

`BLS_SIG_BLS12381G2_XMD:SHA-256_SSWU_RO_NUL_`

(the final `NUL` marks the basic scheme).

The aggregate. The signature the chain carries is

$$\sigma_{\text{agg}} = \sigma_0 + \sigma_1 + \cdots + \sigma_N \in \mathbb{G}_2, \quad (4)$$

of constant size 96 bytes regardless of N . It is accumulated along the chain: each participant receives the running aggregate from its predecessor, adds its own contribution, and forwards the sum. This does not keep the individual contributions secret — an adversary on the network (Section 3.2) observes successive partial sums and recovers any σ_k by subtraction — and nothing in the security argument requires that it should: Theorem 3 rests on the digest recursion, not on the secrecy of the summands. Line 49’s `VERIFYCHAIN` is the multi-pairing

$$e(g_1, \sigma_{\text{agg}}) \stackrel{?}{=} \prod_{k=0}^N e(pk_k, \text{HashToG}_2(m_k)). \quad (5)$$

Distinct messages. The basic scheme’s aggregate verification rejects any aggregate in which two messages are equal, and its security for distinct messages is that of Boneh, Gentry, et al. (2003), which does not depend on how keys were registered. Line 48 establishes the precondition, and the digest recursion guarantees it by construction. Participants still register their keys with proof-of-possession (Section 5.3), for attribution rather than for aggregate security.

Point validation. Group elements use the standard ZCash-format compressed serialization for BLS12-381, and decoders perform subgroup membership checks on every group element to defend against small-subgroup attacks. This check is non-trivial on BLS12-381 and has been the source of real-world implementation vulnerabilities; implementations should use library-provided subgroup checks (e.g., `blst_p1_in_g1`) rather than reimplementing them, and registries apply the same check to registered public keys. The pairing-friendly-curves draft asks protocols adopting its serialization to state three further choices (Sakemi et al., 2026). Only the compressed form is accepted, so each point has exactly one encoding. The identity element is rejected wherever a public key or signature is decoded: for public keys this is the key-validation rule of the signature draft (Boneh, Bradley, et al., 2026), and an honest chain produces an identity aggregate only with negligible probability. No scalar appears in any protocol structure, since secret keys never

leave their holders; a zero secret key is excluded at registration, because its public key is the identity.

We retain the variant because its 96-byte signature is the natural design goal for multi-hop chains, and because Section 8 measures what it costs: in every configuration we measured, it is slower than the default, and it saves only a few percent of a chain’s bytes.

5 Identity Infrastructure

The cryptographic protocol of Section 4 assumes that verifiers can resolve identity-to-public-key bindings for all participants in a chain. This section describes the federated identity infrastructure providing those bindings, including registration with proof-of-possession, public-key resolution, key rotation, and revocation.

5.1 Federated Registry Model

DelegationChain does not assume a global root of trust. Each participating organization operates its own *registry*, which issues identity certificates for agents, signing services, and approval services within that organization. Each registry holds a long-lived root keypair, with the root public key serving as the trust anchor for all certificates the registry issues.

Two organizations establish trust through one of three mechanisms, chosen per deployment. *Direct peering* exchanges registry root keys out-of-band as part of a business agreement; this is the simplest model and adequate for small numbers of partners. *Federated transparency* publishes registry root keys and certificate issuances to a shared append-only log, modeled on Certificate Transparency (Laurie et al., 2021) but adapted to DelegationChain’s certificate format; verifiers trust the log’s commitments and obtain inclusion proofs binding claimed keys to log entries. *Hierarchical attestation* introduces an attestation authority signing statements binding registry roots to organizational identifiers, analogous to a root CA. The protocol permits all three; we recommend direct peering or federated transparency over hierarchical attestation, the latter inheriting the trust-concentration concerns of Web PKI.

These mechanisms are not mutually exclusive. A production deployment may use direct peering with frequent partners and a federated log for the long tail. Whatever the mechanism, the peering relationship is also where the receiving organization fixes which of its partner’s policies it will honor. Its verifiers pin those policy hashes and reject sessions naming any other (Section 6).

5.2 Identifiers and Registration

Identifiers are structured as `organization-id:kind:local-id`, where `kind` is one of `agent`, `signer`, `approver`, `issuer` (the issuance service that signs session tokens), or `service` (a tool or MCP server acting as a verifier). The `kind` tag, which certificates also carry, prevents confusion across roles. The verifier enforces it rather than leaving it as a convention: each position in a chain, and each approval receipt, must be signed under a certificate of the matching kind (Section 4.6). A `service` identifier names a verifier and serves only as the audience of an invocation (Section 4.3). A verifier never has to prove its identity in this protocol, so it needs no certificate. The `signer` kind identifies a signing service in its own right, for operational authentication outside this protocol, for example between an agent runtime and the signing service that holds its key. A signing service signs chain bodies only with the agent keys it holds, under those agents’ certificates; no position in a chain, and no receipt, accepts a `signer` certificate. The `organization-id` matches the issuing registry’s identifier.

Registration produces a certificate binding an identifier to a public key, signed by the registry. Certificates contain version information, the identifier, the public key, the kind, the registry’s identifier (`registry_id`) and public key, issuance and expiry timestamps, a not-before timestamp, a serial number, and the registry’s signature. Default certificate lifetimes are short — 24 hours to 7 days for agents and signing services, with longer lifetimes, up to a maximum the registry publishes, permitted only in combination with active revocation (Section 5.6 uses that maximum to bound how long a verifier must retain a revocation). Short lifetimes reduce reliance on revocation by ensuring that compromised credentials naturally expire within a bounded window.

5.3 Proof of Possession

We require proof-of-possession (PoP) at registration for attribution. A PoP shows that the registrant controls the secret key for the public key it asks the registry to certify. Without it, an adversary could register another principal’s public key under its own identifier (T5a). The adversary would gain no ability to sign, but the registry would then certify one key to two parties. A signature verifying under that key would no longer identify who made it, and that identification is what non-repudiation (Section 3.5) rests on.

DelegationChain requires every registration to include a PoP signature over a registry-issued challenge. The challenge includes the registrant’s claimed identifier and public key, the kind tag, the registry’s identifier, a registry-issued fresh nonce (single-use, short expiry), and a timestamp. The registrant signs the challenge’s digest with the secret key corresponding to the public key being registered, and the registry verifies the PoP before issuing the certificate. The registry-issued nonce prevents replay of PoPs captured from other contexts. A registration proof must never be presentable as a chain signature, or the reverse. Under BLS, a domain separation tag distinct from the one used for chain signatures separates the two; under Ed25519, which has no domain separation tags, the challenge digest’s own tag does. Registries also reject a public key that fails the scheme’s point validation: a small-order Ed25519 key, or, under the aggregate variant, a BLS key outside the prime-order subgroup (Section 4.8).

Under the aggregate variant, PoP is often presented as a defense against rogue-key attacks on BLS aggregation, and in the setting where such attacks work — many signers on a common message — it is one; Ristenpart and Yilek (2007) show that a proof of possession as simple as a signature over the certificate request suffices against rogue-key attacks on BLS-based multisignatures. That is not our setting. An adversary who registered a key $pk^* = pk' - \sum_i pk_i$, built from chosen targets pk_i and a key pk' it controls, could forge an aggregate on a message the targets had all signed, and no such message exists in a chain. Every aggregate is over pairwise distinct messages, which is itself the rogue-key defense of the basic scheme (Boneh, Gentry, et al., 2003; Boneh, Bradley, et al., 2026), so chain integrity (Theorem 3) does not depend on PoP in either instantiation.

5.4 Public Key Resolution

Verifiers resolve an identifier, together with the public key that the body being checked names for it, by querying the issuing registry’s resolution endpoint. Among the certificates binding that identifier to that key, resolution returns the most recently issued one whose validity window contains the verification time t , or, if there is none, the most recently issued one. A renewal issued in advance, or one that ends earlier, therefore never hides an older certificate that is still valid. Whether the binding is usable at t — within its certificate’s validity window, and not revoked — is decided by the verifier at line 27 of Algorithm 1, not by the resolver, so line 27 still rejects when no certificate for the binding is valid. During the overlap of a scheduled rotation (Section 5.5) two certificates for one identifier are valid, and the key selects between them; this

applies to issuers, agents and approval services alike. The response includes the certificate and, if the registry is integrated with a transparency log, an inclusion proof. The verifier validates the certificate signature under the root public key it holds for the organization the identifier names, that is, for the identifier’s **organization-id** prefix, bound to a root key by whichever trust mechanism of Section 5.1 the two organizations use. It also requires the certificate’s **registry_id** to equal that **organization-id**, and the certificate’s kind to be the role the identifier is used in: **issuer** for a session token, **agent** for a delegation or invocation, **approver** for a receipt. If an inclusion proof is present, the verifier checks it against the log’s current signed tree head.

This binding between organization identifier and root key carries real weight. A registry is authoritative only for identifiers in its own organization’s namespace. The naming rule of Section 5.2 is enforced here, by the verifier, rather than left as a convention. Without it, a registry the verifier trusts for one organization could issue a certificate for another organization’s identifier: a certificate substitution (T5d). And because the verifier looks up pinned policies by the organization named in the session body (Section 6), whoever controlled that registry could impersonate the other organization’s issuer and grant sessions under the policies pinned for it.

Verifiers cache resolved certificates to avoid registry queries on every verification. Cache entries are valid until the earlier of the certificate’s expiry and a configurable cache TTL (typically one hour). Cache invalidation is event-driven for revocation: observation of a revocation assertion immediately evicts every cached certificate for the revoked binding.

Policy resolution. The verifier must also hold the policy that a session token’s **policy_hash** commits to; Section 4.6 checks the session scope against it using the containment relation of Section 6.4, Section 6.3 then evaluates the invocation against the innermost scope, and Theorem 6 depends on that policy being one the verifier accepted, not one the chain selected. Since the policy is written by the sending organization and enforced by the receiving one, it has to get across the boundary somehow, and the protocol should not leave this unsaid.

Registries host policy documents alongside certificates, retrievable by hash: a verifier holding **policy_hash** h requests the document h names, and accepts it only if $H(\text{Canon}(\text{document})) = h$. Because resolution is content-addressed, the registry is not trusted for integrity — it cannot substitute a different policy, only decline to serve one — which is the same posture the rest of Section 5 takes towards registries. Policies are immutable under their hash, so a document once fetched is cached for as long as the verifier keeps it; a policy change produces a new hash and, hence, a new session token rather than a silent change under an existing one.

Pinning (Section 6) keeps this fetch off the request path. Because a verifier accepts only hashes it has pinned, it knows every hash it will need before any chain names one, and it can fetch and check the documents when it pins them. A hash it has not pinned is rejected without any fetch. A verifier that defers fetching to first use pays one round trip on that call; Section 9.2 discusses what the remaining dependencies cost the offline-verification claim.

5.5 Key Rotation

The protocol supports three rotation modes. *Scheduled rotation* issues a new certificate with overlapping validity windows: during the overlap, both old and new keys are valid, ensuring that chains in flight at the moment of rotation continue to verify. Default rotation cadence is every 30 days for agents and every 90 days for signing services and approval services. Because policies name approval services by identifier rather than by key (Section 4.5), rotating an approval service’s key needs no policy change. *Emergency rotation* revokes the prior binding (Section 5.6) and certifies a new key, without overlap. *Registry root rotation* is the most sensitive operation: the current root signs a rotation attestation binding the new root to the old, which is then

distributed via the same channel used for the original root. Compromised-root recovery requires out-of-band re-bootstrap of trust and cannot be handled by in-band rotation.

Rotation interacts with chains that are in flight when a key changes. Under scheduled rotation the overlapping validity windows cover this case: a chain signed under the previous key continues to verify for the duration of the overlap, because the previous certificate is still valid. Under emergency rotation there is no overlap, and chains signed under the withdrawn key are rejected. The verifier always resolves the certificate valid at *verification* time. It does not attempt to resolve the certificate that was valid when a token was signed, because its only evidence of signing time would be a field in the token body, signed by the very key whose compromise is at issue; Section 5.6 develops this point.

5.6 Revocation

DelegationChain prefers short-lived certificates over revocation infrastructure: with default certificate lifetimes of 24 hours to 7 days, the maximum window during which a compromised key can be misused is bounded by certificate expiry. Explicit revocation is provided as a defense-in-depth mechanism for deployments choosing longer-lived certificates.

A revocation assertion is a signed statement by a registry that a binding — an identifier together with a public key — is revoked. It revokes every certificate for that binding, and the registry must not certify the binding again; the serial number of the certificate that prompted it is carried for audit only. Revoking by binding rather than by certificate matters because one binding can have several unexpired certificates, for example after a same-key renewal, and revoking only one of them would leave the key accepted under another. A key bound to several identifiers must be revoked under each. A verifier accepts an assertion only from the registry whose namespace contains the identifier it names, as for certificates (Section 5.4). A verifier retains a revoked binding until every certificate for it has expired: for the registry’s maximum certificate lifetime after the revocation, plus a bound on clock skew. The skew bound is set per deployment; it bounds the same quantity as the clock-skew tolerance in the nonce cache’s TTL (Section 4.6), and the reference implementation uses one value, 60 seconds, for both. Verifiers learn of assertions through polling (default interval 60 seconds), push notification, or transparency log monitoring; the choice is a deployment decision. The end-to-end window between key compromise and unforgeability is bounded by the sum of detection latency, reporting latency, assertion issuance, propagation, and residual cache time, of which only the propagation and cache terms are bounded by configuration (typically below 120 seconds). Detection and reporting latency are unbounded by the protocol: they depend on the deploying organization’s monitoring, and nothing in the design constrains them. The property claimed in Section 3.5 is accordingly bounded revocation *propagation*, not bounded exposure.

Revocation applies relative to *verification time*. A binding revoked at time t_r is not accepted by any verifier that has learned of the revocation, including for chains whose tokens were signed before t_r . The gentler alternative — continuing to honor chains signed before t_r , so that revocation does not retroactively invalidate work already in progress — is not available to us. A verifier’s only evidence of signing time is a timestamp in the token body, and that timestamp is signed by the key whose compromise prompted the revocation. An adversary holding a stolen key, observing the revocation, can set the field to any earlier value and produce a chain that verifies against the pre-revocation certificate. A revocation rule resting on that field can be defeated by backdating, which would leave the mechanism without force.

We therefore accept the operational cost. Chains in flight when a revocation is issued may fail verification and must be reconstructed under a rotated key. Two things bound the exposure: only chains outstanding within the propagation window are affected, and scheduled rotation (Section 5.5) avoids the situation entirely by overlapping the validity windows of the old and new

certificates, so that routine key changes never invalidate work in flight. Emergency rotation, by construction, does.

6 Policy Expression Language

This section introduces the policy expression language used to describe agent authorization scopes throughout DelegationChain. The language serves two roles, and separating them is worth doing at the outset, because the verifier treats them differently.

A *policy* is a scope expression published by an organization for its agents and committed to by the `policy_hash` field of the session token (Section 4.3). It is a ceiling: the outer bound on what any agent operating under that session may do, fixed by the organizations involved rather than by any agent. The session names which policy applies; the verifier decides which policies it will accept at all, as described below, and resolves the document from the registry by hash (Section 5.4). A *scope* is a scope expression carried inside a token — the authority the issuer granted the orchestrator, in the session token, and the attenuated sub-scope each delegator passed on, in a delegation token.

Both are drawn from the same grammar, and a single relation connects them. A chain is accepted only if

$$S \supseteq \text{scope}(B_0) \supseteq \text{scope}(B_1) \supseteq \cdots \supseteq \text{scope}(B_{N-1}), \quad (6)$$

where S is the resolved policy and $B_0, \dots, B_{N-1}$ are the session and delegation bodies. Every link is checked by the containment procedure of Section 6.4; the verifier then evaluates the invocation against $\text{scope}(B_{N-1})$, the innermost scope.

The first link needs care, because the issuer signs both `policy_hash` and $\text{scope}(B_0)$ in the same body. If the verifier accepted whatever hash a session named, the check would compare the issuer against a policy the issuer had chosen. Content addressing prevents a different document from being substituted behind a hash, but it does nothing about which hash is selected. The verifier therefore accepts only policies it has *pinned* for the issuing organization: a set of policy hashes fixed in its own configuration, or in the peering agreement with that partner (Section 5.1). A session naming any other hash is rejected. The ceiling S is then drawn from a set the receiving organization agreed to, and the issuer can do no more than choose among policies the verifier has already accepted.

Pinning also closes a gap the protocol would otherwise leave: the receiving organization had no way to state its own terms, only to enforce the sending organization's. A pinned policy may be one the sending organization published and the receiving organization reviewed, or one the two wrote together for the relationship. Either way, it is never one the receiving organization has not seen. What pinning does not defend against is the verifier's own configuration. We treat the pinned set as part of the verifier's trust base (Section 7.1).

With pinning in place, every link of Eq. (6) belongs to the scope monotonicity property of Section 3.5, and Theorem 4 states it for the whole chain.

One consequence carries the weight of Theorem 6. Definition 1 requires a contained scope's approval requirement to be at least as restrictive as its container's, so approval requirements can only strengthen as one moves right along Eq. (6). An approval requirement written into the policy therefore cannot be weakened by the issuer, nor by any agent further down the chain. This is what lets the verifier evaluate against $\text{scope}(B_{N-1})$ — data the agents wrote — and still be certain the requirement it obtains is no weaker than the one in a policy it has itself accepted.

The design of the policy language is constrained by three requirements arising from its role in the protocol. First, the verifier must be able to check cheaply, given two scope expressions S_1 and S_2 purportedly in a delegation relationship, whether $S_2 \subseteq S_1$ — the scope monotonicity

check of Section 4.6 — erring towards rejection where it cannot establish containment. Second, evaluation of an invocation against a scope must terminate in time bounded by the size of the scope expression and the invocation, with no possibility of unbounded computation. Third, scope expressions must be free of side effects: evaluation must not perform network calls, file I/O, or other operations that could be exploited for information exfiltration or denial of service.

These requirements rule out general-purpose policy languages such as Rego or Cedar, which are powerful but operationally expensive to reason about and historically have included features (e.g., HTTP fetches in some versions of Rego) that violate the side-effect-free requirement. DelegationChain’s policy language is deliberately restricted to a fragment that supports realistic enterprise authorization conditions while admitting straightforward decidability and termination analyses.

6.1 Grammar

The grammar of scope expressions, in Backus-Naur form:

```

scope    ::= rule+ | "allow" "all" | "deny" "all"
rule     ::= "allow" at tool action params? where? approval?
at       ::= "at" "=" principal
tool     ::= "tool" "=" identifier
action   ::= "action" "=" identifier
params   ::= "params" "{" binding ("," binding)* "}"
binding  ::= path ":" type
where    ::= "where" expr
expr     ::= expr "and" expr
          | "(" expr ")"
          | atom
atom     ::= path numop value
          | path strop string-value
          | path "in" list-value
          | path "under" string-value
approval ::= "approval" "requires" principal ("," principal)*
type     ::= "int" | "string" | "bool"
numop    ::= "<" | "<=" | "==" | ">=" | ">"
strop    ::= "==" | "starts_with" | "ends_with" | "contains"
path     ::= identifier ( "." identifier)*
principal ::= identifier ":" identifier ":" identifier
identifier ::= letter (letter | digit | "_" | "-")*
value    ::= integer | string | boolean
list-value ::= "[" value ("," value)* "]"
string-value ::= string

```

A scope is either a special form (**allow all**, **deny all**) or a non-empty disjunction of *rules*. Each rule specifies the verifier it authorizes (**at**), a tool, an action, an optional declaration of expected parameter types (**params**), an optional condition on parameter values (**where**), and an optional approval requirement (**approval**). A *principal* is a registry identifier of the form **organization-id:kind:local-id** (Section 5.2); its kind must be **service** in an **at** clause and **approver** in an **approval** clause, and a policy that breaks either rule is malformed. The semantics of a scope is the union of the invocations permitted by its rules; **deny all** permits nothing, and **allow all** is a reserved form used for top-level trusted contexts (e.g., the issuer’s session-issuance authority). It permits every audience as well as every tool and action, so the mandatory audience clause does not constrain it. A verifier that pins **allow all** for a partner accepts that partner’s sessions for anything addressed to it, with no policy limiting them, so it is not meant to be pinned across organizations.

The condition language inside **where** clauses is *conjunctive*: a clause is a conjunction of

atomic predicates over parameter values. Each atomic predicate is one of four forms — numeric comparison, string operator, list membership (**in**), or path-prefix containment (**under**). No predicate involves function calls, network operations, or unbounded iteration.

The three parameter types are disjoint at the value level. Each corresponds to a different CBOR major type in the parameter encoding of Section 4.4 — integers, text strings, booleans — so every parameter value has at most one type, and a leaf value of any other encoded type (a byte string, null or an array) matches no declaration. The disjointness matters. If two types shared values, two rules declaring the same parameter under different types could both match one invocation, and the containment procedure of Section 6.4 would be unsound; a type for filesystem paths, whose values are text strings, would do exactly that. Path-prefix containment is therefore an operator on strings rather than a type. **p under q** holds when the value of string parameter **p** is a canonical absolute path and the segments of **q** are a prefix of its segments, equality included. A canonical absolute path starts with **/**; it is either **/** alone or **/** followed by non-empty segments joined by **/**; no segment is **.** or **..**; and there is no trailing **/**. A non-canonical value makes the atom false. The canonical-form condition is not optional: without it, **/home/agent/workspace/../../etc/passwd** would pass a check for **under "/home/agent/workspace"**. The literal **q** must itself be a canonical absolute path. A policy containing a non-canonical one is malformed and rejected when it is loaded, since such a prefix would otherwise silently match nothing.

Lexical details. Whitespace is insignificant. Strings are double-quoted, with JSON escapes. Integers are decimal, with an optional leading **-**, and lie in the CBOR integer range $[-2^{64}, 2^{64} - 1]$. Booleans are **true** and **false**. In identifiers, **letter** is **[A-Za-z]** and **digit** is **[0-9]**, which rules out Unicode confusables and makes normalization irrelevant for identifiers. String literals are NFC-normalized, like the text values they are compared with (Section 4.4), and the string operators compare byte-wise.

Well-formedness. The grammar admits scopes that make no sense, and the procedures of Sections 6.3 and 6.4 are defined only for well-formed ones. A rule is *well-formed* when three conditions hold.

1. Every path its **where** clause mentions is declared in its **params**. No path is declared twice, and no declared path is a strict prefix of another (**a** together with **a.b**).
2. Every atom pairs its operator, the declared type of its path and its operand as in the table below. Any other combination is malformed.
3. The principal in its **at** clause has kind **service**, and those in its **approval** clause have kind **approver**.

Operator	Path type	Operand
< <= >= >	int	integer
==	any	a literal of the path's type
starts_with ends_with contains	string	string
in	any	a non-empty list, every element of the path's type
under	string	a canonical absolute path

The operator **==** appears in both **numop** and **strop**; the declared type of the path decides which is meant. A scope is well-formed when all of its rules are, and a scope that violates any of these conditions is *malformed*. **LOADPOLICY** rejects a malformed policy. A malformed scope inside a session or delegation body makes that body malformed, and the chain is rejected when it is

decoded, at line 2 of Algorithm 1. The containment procedure of Section 6.4 also returns false on a malformed scope, as a defensive check that a decoded chain never reaches. Implementations also bound the size of a scope — its rules, parameters, atoms, list lengths, text lengths and path depth — so that evaluation and containment stay bounded on input the sender chooses; those bounds belong in a normative specification. One consequence is used in Proposition 2. In a well-formed rule, every path the clause mentions is declared, so it is present in any invocation that passes the rule’s closed-world check, and step 1(c) of the evaluator cannot fail.

Disjunction and negation are deliberately absent from the expression language. Disjunction is recovered one level up: a scope is a union of rules, so a condition $a \vee b$ is written as two rules, and the expressive loss is close to nil. Negation is recovered only in part. Numeric comparisons have complements within the language — the negation of $x > c$ is $x \leq c$ — but the string, membership, and path-prefix predicates do not, so a condition such as “the path is *not* under /etc” cannot be written. We accept this. Authorization policies are more naturally written as enumerations of what is permitted than as exclusions of what is not, and the restriction is what buys the containment procedure of Section 6.4 its polynomial running time: implication between arbitrary propositional formulas is coNP-complete, and a verifier is not a good place to run a SAT solver on an input the sender chose.

6.2 Example

A representative policy fragment governing a payments tool:

```
allow at=orgb:service:payments
  tool=payments action=transfer
  params { amount: int,
           to_account: string }
  where amount <= 1000
allow at=orgb:service:payments
  tool=payments action=transfer
  params { amount: int,
           to_account: string }
  where amount <= 100000
    and to_account in
      ["acct_vendor_a", "acct_vendor_b"]
  approval requires orga:approver:finance
allow at=orgb:service:files
  tool=files action=read
  params { file: string }
  where file under "/home/agent/workspace"
```

This policy permits, at two named services of Organization B, unconditional transfers up to 1,000 units; transfers up to 100,000 units to two approved vendors with finance approval; and read access to files anywhere under a specific workspace directory. Any invocation falling outside these three rules is denied by the implicit closing rule `deny all`.

6.3 Evaluation Semantics

Given a scope S and an invocation $I = (\text{aud}, \text{tool}, \text{action}, \text{params})$, whose audience is the `aud` field of the invocation body, the evaluator returns `ALLOW`, `ALLOW-WITH-APPROVAL(services)`, or `DENY`. The special forms are decided directly: `allow all` returns `ALLOW` and `deny all` returns `DENY`. Otherwise S is a list of rules, and the procedure is:

1. For each rule r in S in declaration order:
 - (a) If r ’s audience, tool, or action does not match I , continue.

- (b) Type-check $I.\text{params}$ against r 's **params** declaration. The check is *closed-world*: the rule matches only if $\text{dom}(I.\text{params})$ is exactly the set of parameter paths r declares, and every value has its declared type. If a declared parameter is missing, if any parameter has the wrong type, or if the invocation carries a parameter r does not declare, continue.
 - (c) Resolve each path mentioned in r 's **where** clause against $I.\text{params}$. If any path resolves to absent, continue (treating the rule as inapplicable).
 - (d) Evaluate r 's **where** clause. If false, continue.
 - (e) If r has no approval requirement, return ALLOW. Otherwise return ALLOW-WITH-APPROVAL(r 's required approval services).
2. If no rule matched, return DENY.

The closed-world condition. Step 1(b) is closed-world by design: an invocation carrying a parameter that the rule does not declare matches no rule, and is denied. The alternative — checking declared parameters and ignoring the rest — would leave undeclared parameters cryptographically bound but never authorized. Under Section 4.4 every supplied value sits inside $\text{Canon}(P)$ and is therefore signed and non-repudiable, but binding is not constraint. A rule declaring **amount** and **to_account** says nothing about a third parameter **override_limits**, and a verifier that accepted an invocation carrying it would be honoring an authorization the policy never granted. Closing the world is what makes a parameter declaration a constraint rather than a schema hint.

The audience clause. Every rule names the verifier it authorizes, and step 1(a) requires that name to equal the invocation's **aud**. The clause is mandatory for the same reason the parameter check is closed-world. The **aud** field of Section 4.3 binds an invocation to one verifier, but binding is not constraint. Without the clause, **tool=payments** would authorize payments at every organization that pinned the policy. A prompt-injected agent could then address a fresh, correctly signed chain to the wrong one and remain within policy. A policy used with several partners therefore lists its rules per verifier.

Match order. Rules are tried in declaration order and the first match wins, so a broad early rule silently suppresses the approval requirement of a narrower rule later in the scope. In the policy of Section 6.2 a 500-unit transfer to **acct_vendor_a** matches rule 1 and returns ALLOW without ever reaching rule 2's finance approval. This is a policy-authoring hazard rather than a protocol defect — the outcome is exactly what the written policy says — but it is easy to introduce by accident, and authors should order rules from most restrictive to least, or scope early rules narrowly enough that they cannot shadow a later approval requirement.

The evaluator is deterministic, has no side effects, and contains no loops over data of unbounded size: the outer loop iterates over rules (a finite syntactic component of S), and **where** clause evaluation is a finite recursive descent over the expression tree.

Proposition 1 (Termination). *Evaluation of an invocation against a scope S terminates in time $O(|S| \cdot |\text{params}|)$, where $|S|$ is the syntactic size of the scope expression and $|\text{params}|$ is the total size of the invocation's parameter encoding.*

Proof sketch. The outer loop iterates over rules of S , of which there are at most $|S|$. For each rule, parameter resolution traverses the parameter map, which is bounded by $|\text{params}|$; **where** expression evaluation is a recursive descent whose depth is bounded by the syntactic depth of the expression, itself bounded by $|S|$. The total work is bounded by the product, yielding the claimed bound. $\square$

6.4 Subset Relation

The verifier of Section 4.6 must check that the sub-scope at each delegation hop is a subset of its parent’s scope. This is a check over scope expressions, performed once per hop at verification time, and not a runtime check over invocations. We define the relation semantically and then give a syntactic procedure that soundly under-approximates it.

Definition 1 (Subset relation). Given scope expressions S_1, S_2 , we say $S_2 \subseteq S_1$ iff every invocation I that the evaluator accepts under S_2 is also accepted under S_1 , with at least as restrictive an approval requirement.

Containment is checked — conservatively, as Proposition 2 and Remark 1 make precise — by the following procedure, which first returns false if either scope is malformed (Section 6.1):

1. If S_1 is `allow all`, return true.
2. If S_2 is `deny all`, return true. If S_2 is `allow all` (and, by step 1, S_1 is not), return false.
3. For each rule r_2 in S_2 :
 - (a) Find the first rule r_1 of S_1 , in declaration order, such that $r_2 \subseteq r_1$ as rules. If there is none, return false.
 - (b) Consider every rule r' that precedes r_1 in S_1 , agrees with r_2 on audience, tool, action, and parameter declaration, and has a **where** clause jointly satisfiable with r_2 ’s. Skip r' if some rule r'_2 preceding r_2 in S_2 agrees with r' on audience, tool, action and declaration and has a clause implied by r' ’s. Otherwise check that r' ’s approval requirement is no stronger than r_2 ’s, and return false if it is stronger.
4. Otherwise, return true.

Step 3(b) is needed because rules are matched in declaration order (Section 6.3). An invocation accepted under S_2 is evaluated under S_1 against the *first* rule of S_1 that matches it. Since r_1 matches every invocation r_2 does, that first rule is r_1 or one of the rules before it. Rules after r_1 never decide such an invocation, and step 3(b) does not examine them. Without the check, a child scope could quietly drop an approval requirement imposed by an earlier parent rule — exactly the escalation Definition 1 is meant to exclude — and the procedure would not notice.

The skip condition handles the converse case, in which a rule of S_2 is itself shadowed. An invocation that r_2 decides under S_2 matched no earlier rule of S_2 , and in particular did not match r'_2 . Suppose it matched r' . It would then pass r'_2 ’s closed-world check, since the declarations are identical; every path in r'_2 ’s clause would be present, since in a well-formed rule those paths are declared; and it would satisfy that clause, which r' ’s implies. So it would match r'_2 , which it did not. Hence r' cannot be the rule that decides it under S_1 . Without the skip, the procedure would reject policies written in the order Section 6.3 recommends. With an approval-requiring rule placed before an overlapping permissive one, such a policy would fail to contain itself. Phase 6 would then reject every session whose scope is the pinned policy, and every agent that re-delegates its whole scope unchanged.

With both conditions the procedure is reflexive: $P \subseteq P$ for every scope P . For each rule r_2 of P , step 3(a) succeeds, since r_2 subsumes itself, and the r_1 it finds lies at or before r_2 . Every rule r' preceding r_1 is therefore an earlier rule of P itself, and it is skipped with $r'_2 = r'$. The incompleteness of Remark 1, which concerns rules covered only by unions of other rules, remains.

Rule subsumption ($r_2 \subseteq r_1$) requires: the audiences, tools and actions match; r_1 and r_2 declare the *same* set of parameter paths with the same types; r_1 ’s approval requirement is no stronger than r_2 ’s (so that any approvals satisfying r_2 also satisfy r_1); and r_2 ’s **where** clause implies r_1 ’s.

The identity condition on parameter declarations follows from the closed-world rule of Section 6.3 together with the disjointness of types (Section 6.1). A rule matches only invocations whose parameter set is exactly what it declares and whose values have exactly the declared types, and each value has at most one type. The declaration of any rule matching an invocation I is therefore determined by I alone, and two rules with different declarations — different parameter sets, or the same parameters under different types — accept disjoint sets of invocations: declaring fewer parameters is strictly more restrictive rather than more permissive, and no containment holds in either direction. Requiring identical declarations is therefore the only sound relation available without further machinery, such as an explicit notion of optional parameters that the language does not currently have.

The remaining sub-problem is deciding implication between **where** clauses. Because a clause is a conjunction of atoms (Section 6.1), this reduces to a check over atoms rather than to satisfiability. The atoms have a restricted structure: numeric comparisons are over a fixed integer path and constants, string operators are over fixed string paths and constant patterns, list membership is over fixed lists, and path-prefix containment is over fixed prefixes. Atoms over distinct paths are independent, so a conjunction $C_2 = \bigwedge_j a_j$ implies a conjunction $C_1 = \bigwedge_i b_i$ exactly when, for each b_i , those atoms of C_2 over the same path as b_i jointly imply b_i . Joint satisfiability, needed by step 3(b), decomposes the same way: a conjunction is unsatisfiable exactly when the constraints induced on some path are inconsistent.

The per-path checks are exact for integer and boolean paths, and for string paths whose value is confined to a finite set by **==** or **in**. There the atoms induce an interval, a finite set, or both, and implication and satisfiability are decided by inspecting them. For string paths without such a set — constraints that combine **starts_with**, **ends_with**, **contains** and **under** — we do not have an exact procedure, and the checks are sound but incomplete. Implication is reported only when a single constraint already forces the atom (a longer required prefix implies a shorter one, for instance). A conjunction is reported unsatisfiable only when two of its constraints are directly incompatible (two required prefixes, neither a prefix of the other, for instance). Each check therefore errs only in the direction containment can tolerate. A missed implication stops step 3(a) or the skip condition from applying, and a missed contradiction makes step 3(b) examine a rule it could have ignored; either way the result can only be a rejection. An atom on a path the other clause does not constrain is judged against the unconstrained domain of the path's declared type. That is sound because, by well-formedness, every path a clause mentions is declared, and the rules being compared have identical declarations. No search over truth assignments is required at any point.

Proposition 2 (Soundness and cost of the containment procedure). *The procedure above terminates in time polynomial in $|S_1| + |S_2|$, and whenever it returns true, $S_2 \subseteq S_1$ in the sense of Definition 1.*

Proof sketch. Cost. The procedure enumerates pairs $(r_2, r_1) \in S_2 \times S_1$, of which there are at most $|S_1| \cdot |S_2|$. Each pair check compares two identifiers, compares two parameter declarations for equality, compares two approval requirements, and decides implication between two conjunctions of atoms. The last of these groups the atoms of each clause by path and, for each atom of r_1 's clause, evaluates an implication check against the constraint that r_2 's atoms induce on the same path. The number of such checks is bounded by the product of the two atom counts, and each takes time polynomial in the sizes of the atoms involved: linear in the literals for the string checks, and in the list lengths for membership. Step 3(b) performs a satisfiability check of the same form over the same pairs, and its skip condition adds, for each r_2 , at most one implication check per pair of a rule of S_1 and a rule of S_2 . The total is polynomial in $|S_1| + |S_2|$.

Soundness. Steps 1 and 2 return true only when S_1 accepts every invocation with no approval requirement or S_2 accepts none, and containment then holds trivially. Otherwise S_2 is a list

of rules. Suppose the procedure returns true, and let I be an invocation the evaluator accepts under S_2 , decided there by the first rule $r_2 \in S_2$ that matches it. Step 3(a) exhibited the first $r_1 \in S_1$ with $r_2 \subseteq r_1$. Then r_1 matches I as well: the audience, tool and action agree, so step 1(a) of Section 6.3 passes; the parameter declarations are identical, so the closed-world check of step 1(b) passes; r_1 is well-formed (Section 6.1), so every path its clause mentions is declared and therefore present, and step 1(c) passes; and r_2 's clause holds of I and implies r_1 's, so step 1(d) passes. Hence I is accepted under S_1 . For the approval component, the rule that actually decides I under S_1 is the first rule of S_1 matching I . Because r_1 matches I , that rule is either r_1 itself, whose approval requirement step 3(a) compared with r_2 's, or a rule r' preceding r_1 . In the second case r' and r_2 both match I , so by the identity argument above they have identical declarations. They also agree on audience, tool and action, and I satisfies both clauses, so those clauses are jointly satisfiable. Nor was r' skipped. A skipping rule r'_2 would precede r_2 in S_2 , agree with r' on audience, tool, action and declaration, and have a clause implied by r' 's. So I , which matches r' , would match r'_2 too — it passes the same closed-world check, every path in r'_2 's clause is declared by well-formedness and so present, and that clause is implied by r' 's — contradicting the choice of r_2 as the first rule of S_2 matching I . Step 3(b) therefore examined r' , and its approval requirement is no stronger than r_2 's. Thus I is accepted under S_1 with an approval requirement no stronger than the one S_2 imposes, which is Definition 1. $\square$

Measured, CONTAINS takes 9.0 μ s in the worst case for scopes of 4 rules with 4 atoms each, and 5146 μ s at 64 rules with 8 atoms each (Section 8.4). Because the procedure's cost grows with the product of the two scopes' sizes, a verifier should cache containment results by scope digest, as the prefix cache of Section 8.4 does.

Remark 1 (Incompleteness). The procedure is sound but not complete. Step 3(a) requires each rule of S_2 to be subsumed by a *single* rule of S_1 , so a child rule covered only by the union of several parent rules is rejected. Take S_1 to be the two rules `allow at=v tool=t action=a params {x: int} where x <= 10` and `allow at=v tool=t action=a params {x: int} where x > 10`, and S_2 the single rule `allow at=v tool=t action=a params {x: int}`. Every invocation accepted under S_2 is accepted under S_1 , so $S_2 \subseteq S_1$ holds semantically; but neither rule of S_1 subsumes S_2 's rule on its own, and the procedure returns false. The incompleteness is conservative — the procedure rejects some legitimate delegations and accepts no illegitimate ones — and we prefer that direction of error. Deciding Definition 1 exactly would require reasoning about unions of rules, which the per-rule structure of the procedure does not support; we leave it open whether the exact relation admits an efficient decision procedure for this language. String constraints are a second, independent source of incompleteness: for string paths without a finite set of admissible values, the per-path implication and satisfiability checks are themselves incomplete (Section 6.4), in the same conservative direction. Two further sources come from the shape of the procedure. First, a child rule that can never decide an invocation — because its clause is unsatisfiable, or because earlier child rules catch everything it matches — must still be subsumed by some parent rule. It is rejected if no parent rule with the same audience, tool, action and declaration has an approval requirement weak enough, although the rule is harmless. Second, the skip test of step 3(b) asks that the whole of r' 's clause imply the clause of a single earlier child rule. It would be enough that every invocation matched by both r' and r_2 is decided by earlier child rules taken together. Both are conservative. Skipping child rules with unsatisfiable clauses in step 3(a) would be sound, and is left to future work.

6.5 Limitations of the Policy Language

The policy language is deliberately bounded. It cannot express semantic conditions of the form “is this email message professional?” or “is this code review constructive?” that would require external evaluation, typically by another LLM. Such conditions are non-deterministic and admit

no obvious decidability or termination guarantee, and we judge them inappropriate for inclusion in authorization policies that must be cryptographically committed to via `policy_hash`.

The language also cannot express conditions referring to state outside the invocation itself: account balances, current time relative to business hours, counts of prior invocations within a window. Such conditions either require stateful evaluation (which would violate the side-effect-free requirement) or external queries (which would introduce latency and additional trust dependencies). Where stateful authorization is required, we anticipate it being handled at the tool implementation layer, after the protocol’s stateless authorization check has succeeded.

The **under** operator is lexical. Its canonical-form requirement stops `..` traversal, but it cannot see the filesystem: a symbolic link inside the permitted tree can point outside it. A tool that accepts a path authorized by **under** must therefore resolve it without following links out of the tree, and the protocol’s check does not relieve it of that.

The closed-world condition of Section 6.3 carries an operational cost of its own. Adding an optional parameter to a tool is not a backward-compatible change: every rule governing that tool must be updated to declare the new parameter before any invocation may supply it, and until the updated policy has propagated, invocations carrying the new parameter are denied. Because policies are committed to by hash in session tokens, that update is a policy version change rather than an edit in place, and every partner whose verifiers accept the policy must pin the new hash before sessions under it will verify. We consider this the right default, since the failure mode is denial rather than unauthorized acceptance, but deployments whose tool schemas change often should expect policy churn in proportion. The audience clause of Section 6.3 has the same kind of cost. Adding a verifier, such as a new region or a staging instance, means adding rules that name it. That is a new policy version, which every partner pinning the policy must re-pin, including partners the new verifier does not affect. This is the re-pinning that Section 4.5 avoids for approval services by naming them by identifier. For audiences we accept it, since which verifiers a policy authorizes is part of what the partners agreed to. An organization can confine the churn by publishing a separate policy for each partner.

These limitations are deliberate. A more expressive language would provide more flexibility but at the cost of guarantees we consider essential. We discuss possible extensions to semantic conditions via verifiable LLM attestation in Section 9.

7 Security Analysis

This section presents the security argument for DelegationChain. We state the protocol’s security properties as theorems, reduce each to standard cryptographic assumptions or to the properties of the policy language, and sketch proofs. The arguments here are informal in the sense that they have not been mechanized in a proof assistant or verified using symbolic protocol analyzers such as Tamarin or ProVerif. We discuss the planned formalization in Section 9.2.

7.1 Assumptions

The security arguments below rest on the following assumptions:

Signature unforgeability. The signature scheme Σ is existentially unforgeable under chosen-message attack (EUF-CMA). For the default instantiation, Ed25519 with strict verification, Brendel et al. (2021) prove this, and strong unforgeability besides. For the aggregate variant, the BLS scheme (Boneh, Lynn, et al., 2001; Boneh, Bradley, et al., 2026) on BLS12-381 (Sakemi et al., 2026) is EUF-CMA under the co-computational Diffie-Hellman assumption on the curve, in the random oracle model for `HashToG2`.

Aggregate security. For the aggregate variant only (Section 4.8): aggregate signatures over pairwise distinct messages are existentially unforgeable (Boneh, Gentry, et al., 2003). No PPT adversary can produce a tuple $(pk_1, \dots, pk_n, m_1, \dots, m_n, \sigma_{\text{agg}})$ with the m_i pairwise distinct such that the multi-pairing check accepts, unless it has obtained valid individual signatures on each m_i from the holder of sk_i . The result holds under the same assumptions as BLS unforgeability. It places no requirement on how the pk_i were registered, nor on their being distinct. Section 4.3 guarantees distinct messages, and phase 8 of Algorithm 2 checks it.

Collision resistance of SHA-256. SHA-256 is collision-resistant in the standard sense.

Honest signing key custody. Signing keys remain in their designated hardware security modules and are not exfiltrated. Compromise of an HSM is explicitly out of scope per Section 3.4.

Honest registry root. Registry root signing keys are honestly managed. Compromise of a root is addressed via rotation and detection (Section 5.5); the formal arguments below assume an honest root, with the implication that root compromise reduces security to a recovery problem rather than a forgery problem.

Verifier configuration integrity. The set of policy hashes a verifier has pinned for each partner, and the root key it associates with each partner organization, reflect its own organization’s decisions and cannot be written by the adversary. The first link of Eq. (6), and through it Theorems 4 and 6, is only as good as that configuration.

7.2 Theorems

Theorem 1 (Token unforgeability). *Under the unforgeability of Σ , no PPT adversary can produce a session, delegation, or invocation token that verifies under a public key pk for which it does not hold the corresponding sk , except with negligible probability.*

Proof sketch. A valid token consists of a body and a signature on the body’s domain-tagged digest. Forging the signature without sk contradicts the unforgeability of Σ under chosen-message attack: any adversary that forges a token can be used to construct an adversary against Σ by treating the domain-tagged digest as the message and extracting the signature. $\square$

Theorem 2 (Parameter binding). *A valid invocation token for parameters P cannot be repurposed as a valid invocation token for parameters $P' \neq P$, except with negligible probability.*

Proof sketch. The signed message m_N contains $\text{Canon}(\text{InvocationBody})$, which includes $\text{Canon}(P)$. Under collision resistance of SHA-256 and deterministic encoding, $\text{Canon}(P) \neq \text{Canon}(P')$ implies $m_N \neq m'_N$. The signature σ_N valid on m_N is invalid on m'_N by the unforgeability of Σ . The redundant `params_hash` field provides defense-in-depth against canonical encoding bugs but is not required for the theorem. $\square$

Theorem 3 (Chain integrity). *The signatures over a delegation chain cover the entire chain content: any modification, reordering, truncation, insertion, or extension of the chain causes `VERIFYCHAIN` to reject, except with negligible probability.*

Proof sketch. Per-hop case. Modifying any body changes its m_k and, because each digest commits to its predecessor, every later digest. Each σ_j signs m_j , so a verifier recomputing the digests from the received bodies sees at least one m_j that its holder never signed, and accepting it contradicts EUF-CMA. The same argument covers reordering, truncation and insertion: each changes the sequence of digests the verifier recomputes, so some recomputed m_j was never signed under the

key that $\text{spk}(B_j)$ names. Extension requires a new signature for the added hop, which requires that hop's secret key.

Aggregate variant. Modification of any body changes its canonical encoding, changing the corresponding m_k (since m_k hashes the body). Reordering changes the sequence of m_k values (since m_k commits to m_{k-1}). Truncation is caught by the same recursion. Suppose the adversary drops B_k for some interior $0 < k < N$. The verifier recomputes the remaining messages from the bodies it actually receives, so where A_{k+1} signed $m_{k+1} = H(\text{TAG_DEL} \parallel m_k \parallel \text{Canon}(B_{k+1}))$ the verifier computes $m'_{k+1} = H(\text{TAG_DEL} \parallel m_{k-1} \parallel \text{Canon}(B_{k+1}))$. Since $m_k \neq m_{k-1}$ except with negligible probability under collision resistance, $m'_{k+1} \neq m_{k+1}$, and every subsequent message differs likewise. The adversary would need signatures under the surviving hops' keys on messages those hops never signed, which contradicts aggregate security. Dropping the first or last hop is caught earlier still, by the position-determined tag check: a chain whose leading token is not a session token, or whose trailing token is not an invocation token, is rejected in phase 1 of Algorithm 1. No assumption about the secrecy of individual σ_k values is required, and we make none — the accumulation scheme of Section 4.8 would not support one. Extension requires producing a new σ for the extension hop; under aggregate security, this requires the secret key of the extending party.

Both. The position-determined tag check (Section 4.3) rejects chains where an invocation token is repurposed as a delegation token (or vice versa) for chain extension. $\square$

Theorem 4 (Scope monotonicity). *In any chain accepted by the verifier, every link of the containment chain Eq. (6) holds in the sense of Definition 1, where S is a policy the verifier has pinned for the issuing organization. In particular, the session scope is contained in S , and the scope at hop $k + 1$ is contained in the scope at hop k .*

Proof sketch. Phase 6 of Algorithm 2 rejects any session whose policy hash is not pinned for the issuing organization. That organization is $\text{org}(\text{sid}(B_0))$, read from the signed session body. Phase 5 of Algorithm 1 has already required the issuer's certificate to verify under the root the verifier holds for that organization, and to name it as its registry. So the pins consulted are those of the organization that actually issued the session, and a registry trusted only for some other organization cannot redirect the lookup. Phase 5 has also required that certificate to be of kind **issuer**. So the session scope is one the organization's issuance service granted, not one an agent or approver granted itself. Phase 6 then resolves S by content address, so by collision resistance of SHA-256 S is the document the pinned hash names. Finally it checks each link of Eq. (6) explicitly, using the containment procedure of Section 6.4. Acceptance of the chain implies that every one of these checks returned true, and by the soundness half of Proposition 2 that implies containment in the sense of Definition 1. The converse does not hold: by Remark 1 the verifier will also reject some chains whose scopes are genuinely nested, so the guarantee is that accepted chains are monotone, not that all monotone chains are accepted. The integrity of the body content from which the policy hash and the scopes are read is guaranteed by Theorem 3. The first link is only as strong as the verifier's pinned set, which is an assumption (Section 7.1) and not something the protocol establishes. $\square$

Theorem 5 (Replay resistance). *Consider a single logical verifier maintaining a correctly-sized nonce cache with an atomic insert-if-absent operation — one cache shared across all instances that accept chains, rather than an independent cache per replica — that accepts a chain only when inserting its nonce-and-invoker-key pair succeeds. It accepts at most one chain for each such pair within the validity window, however requests are interleaved, and rejects every chain presented after the validity window has expired. Moreover, a chain addressed to one verifier is rejected by every verifier with a different identifier.*

Proof sketch. Acceptance requires a successful insertion of the chain’s (pk_N, nonce) pair. The insertion is atomic, so of any number of chains carrying the same pair, whether presented one after another or concurrently, at most one insertion succeeds while the pair remains in the cache; the phase-3 lookup only rejects some replays earlier. The cache TTL is the validity window plus a tolerance at least as large as the clock disagreement among the instances sharing the cache, so the pair is not evicted while a replay could still pass the temporal checks at any of them, and outside the window the verifier rejects on the expiry check before any signature verification. For the second part, the invocation body names its verifier in `aud`, which is part of m_N . Phase 1 rejects a chain whose `aud` is not the verifier’s own identifier, and altering `aud` changes m_N and invalidates σ_N (or the aggregate) (Theorem 3). Distinct verifiers therefore need no shared cache, provided they have distinct identifiers. $\square$

Theorem 6 (Approval binding). *An approval receipt for invocation I cannot be reused for any invocation $I' \neq I$, and an invocation requiring approval from service s cannot be accepted with approval from service $s' \neq s$, except with negligible probability.*

Proof sketch. The approval body signed by the approval service includes the invocation digest of Section 4.5, a tagged hash of the invocation body without its receipts. Modification of the invocation changes this digest, invalidating the receipt under the unforgeability of Σ . Phase 7 checks each receipt under the key the registry currently certifies for the service it names, with kind `approver`, so a receipt signed by any other key, or under a revoked certificate, is rejected. The required approval service identifiers are named by the policy S . The session names S by hash, but the verifier accepts the hash only if it has pinned it for the issuing organization. S is therefore a policy the receiving organization agreed to, not one the issuer was free to choose. Resolution is content-addressed (Section 5.4), so no other document can be substituted behind the hash.

The requirement then propagates inward. By Theorem 4, every link of the containment chain Eq. (6) holds, and by the approval clause of Definition 1 each containment can only strengthen the approval requirement. The requirement the verifier obtains by evaluating the invocation against $\text{scope}(B_{N-1})$ is therefore no weaker than the one S imposes, even though the intermediate scopes were written by the agents themselves. Routing the approval elsewhere disturbs none of this: the policy hash is fixed at session start and does not vary with the agent’s routing choice, so a misrouted chain arrives carrying the correct hash and the wrong receipt, and phase 7 of Algorithm 2 rejects it because no receipt signed by a required service is present. $\square$

7.3 Threats Addressed

The five prevented threat families of Section 3.3.1 are addressed by the above theorems and the identity infrastructure of Section 5, as follows. Token forgery (T1a) is addressed by Theorem 1, and chain forgery (T1b) by Theorem 3. Parameter substitution (T2a) is addressed by Theorem 2, and scope escalation (T2b) by Theorems 3 and 4 together. Cross-session replay (T3a) and intra-window replay (T3b) are addressed by Theorem 5, under the single-logical-verifier hypothesis stated there; cross-context token reuse (T3c) is addressed by domain separation tags (Section 4.3); and cross-verifier reuse (T3d) is addressed by the audience check of phase 1, the second part of Theorem 5. The mandatory audience clause of every rule (Section 6.3) goes further, and lets a policy state which verifiers it authorizes at all. The prevented part of the approval family (T4b–T4d) is addressed by Theorem 6 together with the policy containment chain of Eq. (6); T4a is bounded rather than prevented and is discussed below. The prevented part of the identity family (T5a, T5c, T5d) is addressed by proof-of-possession (Section 5.3), short-lived certificates and revocation (Section 5.6), and optional transparency log integration. Certificate substitution (T5d) additionally relies on the namespace binding of Section 5.4: a certificate must verify under

the root the verifier holds for the organization its identifier names. Role confusion (T5e) is addressed by the kind checks of phases 5 and 7. They accept only an **issuer** certificate for the session body, only **agent** certificates for later hops, and only **approver** certificates for receipts. Pinning adds an informal bound outside the honest-root assumption. An issuer whose keys have been compromised can still grant sessions only within the policies the verifier has pinned for that issuer’s own organization, and because of the kind check a compromised agent or approver key cannot act as an issuer at all. The qualifier depends on the namespace binding. Without it, a compromise in one organization could draw on the policies pinned for another.

The threats of Section 3.3.2 are bounded, not prevented. A *compromised approval service* (T4a) can issue receipts the human approver never authorized, and no check in Algorithms 1 and 2 can distinguish them from honest ones. The bound comes from the decoupling of Section 4.5: because approval signatures are verified separately from the chain signatures, the compromise cannot extend to chain forgery or scope escalation, and it reaches only those invocations that policy routes through the affected service. A *compromised registry root* (T5b) can certify any identity in its own organization’s namespace. Its reach is bounded by the namespace binding of Section 5.4, which keeps it out of every other organization’s identifiers, and by pinning: the sessions it enables stay within the policies verifiers have pinned for that organization. *Within-policy prompt injection* (T6a) — an adversary inducing an agent to take an action that is within its permitted policy but contrary to the principal’s intent — is not prevented by the protocol. The protocol’s contribution is to *bound the blast radius*: the adversary can cause only invocations that policy already permits and, where policy requires human approval, will surface the action to an approver who has an opportunity to refuse. Tightly-scoped policies and appropriately-required approvals reduce the blast radius; defending against prompt injection itself remains an active research area outside the protocol’s purview. *Confused-deputy delegation* (T6b), where an agent is induced to delegate authority to an adversary, is similarly bounded rather than prevented: the agent cannot delegate more authority than it itself holds (Theorem 4), limiting damage to the agent’s own permitted scope.

7.4 Limitations of the Analysis

We emphasize that the analysis above consists of proof sketches under specified assumptions, not machine-checked or peer-reviewed proofs. Several aspects warrant scrutiny in any subsequent formalization. Theorem 3 does not depend on proof-of-possession. In the per-hop case it rests on each signature’s unforgeability; under the aggregate variant it rests on the messages of a chain being pairwise distinct, which the digest recursion guarantees except with negligible probability and which phase 8 of Algorithm 2 also checks explicitly. Proof-of-possession remains a registration requirement, but a failure to enforce it would affect attribution (T5a), not chain integrity. Theorem 5 depends on operational properties of the nonce cache (correct sizing, crash recovery, distributed consistency across verifier replicas) that are not enforced by the protocol but must be discharged by the verifier implementation. The containment procedure of Section 6.4 is sound but not complete with respect to Definition 1 (Remark 1); its soundness has been argued informally here and would repay mechanization, since it is the step on which Theorem 4 rests. The reference implementation’s security suite (Section 8.2) exercises every threat row against the algorithm line that should reject it. That is evidence that the implementation matches the specification, not a proof that the specification is secure. Symbolic verification of the protocol in Tamarin or ProVerif would confirm or refute these arguments and is planned future work.

8 Implementation and Evaluation

This section describes the reference implementation (Section 8.1) and its correctness evaluation (Section 8.2), then a performance evaluation under an analysis plan frozen before any measurement (Sections 8.3 to 8.5). Exploratory analyses, which the plan did not include, follow separately (Sections 8.6 and 8.7), and the section closes with threats to validity (Section 8.8). Every table and figure in it is generated by the benchmark harness from archived raw data and included unedited.

8.1 Implementation

The reference implementation is a Rust workspace of nine crates: `dc-cbor`, a strict deterministic CBOR codec that rejects every non-canonical encoding; `dc-crypto`, the signature schemes behind a common interface; `dc-types`, the bodies, digests, certificates and receipts; `dc-registry`, registration with proof-of-possession, resolution, revocation and the policy store; `dc-policy`, the policy language of Section 6; `dc-chain`, the issuance, signing and approval services; `dc-verifier`, which implements Algorithms 1 and 2 line by line, with one rejection variant per rejecting line; `dc-baselines`, the comparison arms and the prefix caches; and `dc-bench`, the benchmark harness. The verifier is generic over the signature scheme, so every instantiation runs the same phases 1–7 and differs only at line 49. Unsafe code is denied throughout, except for one operating-system call in the benchmark harness. The measurement-relevant dependencies are pinned: `blst` 0.3.17 (Supranational, 2026), `ed25519-dalek` 2.2.0 (dalek-cryptography, 2025) and `biscuit-auth` 6.0.0 (Eclipse Biscuit Project, 2025), built with `rustc` 1.97.1.

The implementation was built specification first. Every choice the paper left open was recorded as a numbered decision, and every defect found in the paper was logged with the evidence for it. The review that preceded implementation, and the implementation itself, surfaced 16 specification defects, among them a soundness gap in an earlier containment procedure and a revocation gap, all corrected in the version presented here.

8.2 Correctness Evaluation

Security suite. Every threat of Section 3, from T1a to T6b, has at least one test, as do the closed-world rule, the structural checks, key-chain consistency and resolution, and phase ordering. Each test asserts the exact algorithm line that rejects the attack, or acceptance where the threat is only bounded. The suite runs under both instantiations: each runs 52 tests, 50 shared and 2 specific to it (54 distinct), and all passed under both when the suite’s table was generated. Section A lists them, with the concurrent-replay test below, which also runs under both.

Concurrent replay. Theorem 5 was tested with 64 threads presenting the same chain in each of 1,000 rounds, 64,000 verifications in all. Every round accepted exactly one copy. Of the rejected copies, 49,638 were stopped by the phase-3 lookup (line 17) and 13,362 only by the atomic insertion of line 50: without atomicity, each of those would have been accepted.

Cache equivalence. Caches must change the cost of a verification, never its outcome (Section 4.6). For each family of instantiations, a randomized run of 10,000 chains compared the uncached verifier with the warm one and, where they exist, with the prefix-cached one: the default instantiation (per-call, batch and prefix-cached), the aggregate variant (with and without its pairing and prefix caches), and individual BLS signatures. The runs interleave clock advances, pin changes, rotations, same-key renewals (including future-dated and shortening ones), revocations of the newer and of the older certificate of a renewed binding, refused re-certifications, and

Table 1: Differential test of the containment procedure and its per-path checks against an enumerating oracle (seed 56324). Soundness: the procedure never answers true where the oracle answers false. Completeness: the share of the oracle’s true answers that the procedure also finds. These are test statistics, not benchmark results.

Procedure	Inputs	Cases	Soundness violations	Oracle true	Completeness
Contains	all well-formed pairs	108,370	0	51,998	0.9197
	no child rule unsatisfiable			36,188	0.9991
	bool			4,767	0.9379
	int			4,649	0.9269
	mixed			22,542	0.8556
	no atoms			16,003	0.9997
	string, finite set			2,077	0.8960
	string, no finite set			1,960	0.9679
<i>Contains reflexivity: 250,533 checks, 0 failures; malformed scopes rejected: 49,467</i>					
Implies	bool	150,000	0	93,335	1.0000
Implies	int	150,000	0	78,059	1.0000
Implies	string, finite set	65,479	0	53,123	1.0000
Implies	string, no finite set	72,628	0	16,367	0.8381
Unsat	bool	150,000	0	54,708	1.0000
Unsat	int	150,000	0	61,273	1.0000
Unsat	string, finite set	65,479	0	51,099	1.0000
Unsat	string, no finite set	72,628	0	11,843	0.7398

replays. No run found a disagreement in decision or in rejecting line. For the aggregate variant, a separate run of 10,000 randomized chains found that the pairing cache agrees with direct aggregate verification on every one.

Differential oracle for containment. Table 1 compares the containment procedure of Section 6.4 with an enumerating oracle on 150,000 generated scope pairs, of which 108,370 are well-formed; the validator rejected 49,467 malformed scopes. The procedure never answered true where the oracle answered false: there were no soundness violations, and no reflexivity failures in 250,533 checks. Its completeness is 0.9197 overall and 0.9991 when no child rule is unsatisfiable, a source of incompleteness that Remark 1 names: under our generator, unsatisfiable child rules account for 4,138 of the 4,176 missed containments. These rates depend on the generator: half of its pairs are independent, and in the other half the child scope is derived from the parent by mutation, so that containment is often true or nearly so. As evidence that the oracle can find errors, three bugs planted in the procedure were all caught.

Fuzzing. The CBOR decoder was fuzzed for ten minutes, 111,517,984 executions, with no crash; the properties checked were that nothing panics and that re-encoding is a fixed point. AddressSanitizer was not available on the toolchain used, and the decoder contains no unsafe code.

8.3 Performance Method

The question. The plan asked: is BLS aggregation a net benefit for DelegationChain once caching is accounted for? It allowed “no” as an answer, to be reported as such.

Arms. Table 2 lists the arms. A, A-ind, C and C-batch run the same verifier; only line 49 and the scheme of the signatures differ.

Table 2: Benchmark arms, in the terms of this paper.

Arm	Description	Role
C	Default instantiation: $N+1$ Ed25519 signatures, each checked with strict verification	primary
D	As C, with a prefix cache: on a hit, only σ_N is verified	primary (deployment pattern)
A	Aggregate variant (Section 4.8): one BLS aggregate	primary
B	As A, with pairing and prefix caches	primary (deployment pattern)
A-ind	$N+1$ individual BLS signatures	ablation
C-batch	As C, verified with Ed25519 batch verification	ablation; different edge semantics
E	Biscuit (Eclipse Biscuit Project, 2025)	positioning reference, with functional gaps
A-mt	As A, with <code>blst</code> ’s thread pool	supplementary

Workloads and states. Four profiles: *small*, one rule with two parameters and one atom; *medium*, the policy of Section 6.2 verbatim, each hop tightening one bound; *large*, 16 rules with 4 parameters and 3 atoms each, including two approval rules, each placed before the permissive rule it overlaps; and *medium-approval*, which is medium with an invocation that requires a finance receipt. Verifiers were measured in four states: *cold*, with empty caches; *warm*, with certificate and policy caches filled by separate warm-up chains; *warm+prefix*, in which every measured call is a prefix-cache hit; and *prefix-miss*, in which every chain has a new prefix.

Grid, machine and statistics. The grid covers $N \in \{1, 2, 3, 5, 10\}$ in each profile, with medium-approval at $N = 3$ only: 231 latency configurations per run, each with 1,000 warm-up and 10,000 measured verifications (fewer for the injected-latency configurations of the cold path), over three runs in separate processes, each in its own seeded random order. Every timed operation is one verification call on pre-generated bytes, and every measured verification was accepted. The machine is an Apple M4 Max (10 performance and 4 efficiency cores) under macOS 26.5.2, on AC power in High Power mode, with release builds, fat LTO and `target-cpu=native`. We report medians with bootstrap 95% confidence intervals pooled over the runs. Each ratio of the aggregating arm to its non-aggregating counterpart is judged with a $\pm 10\%$ margin of practical significance: *not a net benefit* if the interval’s lower bound exceeds 1.10, *net benefit* if its upper bound is below 0.90, and *no material difference* or *inconclusive* otherwise. A verdict stands only if all three runs’ own ratios fall on the same side of the margin. The pooled intervals are often nearly zero-width, because pooling treats the runs’ samples as independent and the timer is fine-grained, so it is the agreement of the runs that makes the verdicts robust; the figures therefore show the range of the three runs’ medians.

The frozen plan and its deviations. The plan was frozen and tagged before any measurement. Two amendments preceded measurement: a calibration probe, run before and after every configuration, as a second throttling signal, and a text correction. Two deviations followed it, both approved by the author; the first was decided before any latency result was opened. First, the safety valve, which aborts a run with more than 10% of its configurations flagged as throttled, had been implemented per process; it was changed to count per run, as the plan specifies, after a process running the multi-threaded supplementary arm aborted, and that process was redone in full. Second, a 200 ms busy spin was added before every probe, because probes that followed sleep-dominated configurations had measured an idle core’s ramp-up rather than throttling. Twenty-six configurations were flagged, every one by the probe and none by the operating system’s thermal report, and were re-run as the plan requires; the two flagged again use their re-run samples. After the results were reported, one report error was corrected — a

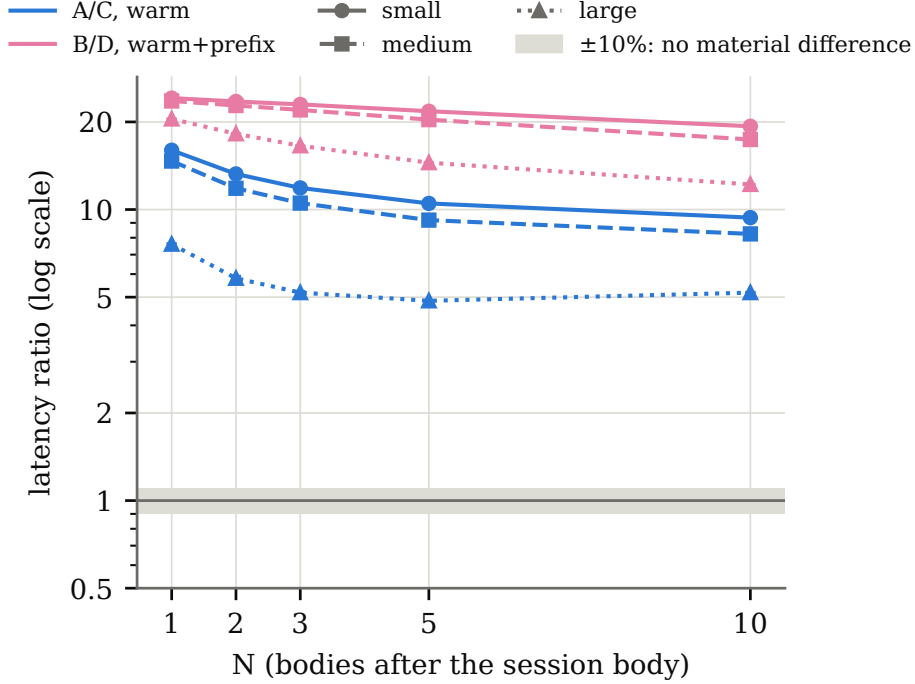


Figure 3: The frozen plan’s verdict ratios: latency of the aggregating arm over its Ed25519 counterpart against N : A/C (warm) and B/D (warm+prefix), in the small, medium and large profiles, log scale. The line marks equal latency, and the band the frozen plan’s $\pm 10\%$ margin of no material difference; every run’s ratio lies above it, so aggregation is not a net benefit at any point. Points are ratios of pooled medians; error bars are the range of the three runs’ ratios of medians.

comparison of token sizes in different units (Section 8.6) — without changing any measurement or verdict. Three attempts to resume measurement were refused by the machine-state check before anything ran.

8.4 Results

Verdicts. Table 3 gives the verdict for every cell. In the deployment pattern the plan asks about — arm B against arm D, warm with a prefix cache, $N = 3$, medium — the latency ratio is 21.995 [21.994, 21.995], and every run agrees (22.003, 21.951, 21.997): aggregation is not a net benefit. It is not a net benefit in any cell, with caching or without. B is 12.2–24.2 $\times$ slower than D, and warm A is 4.9–16.0 $\times$ slower than C and 5.8–19.5 $\times$ slower than C-batch, in all 16 cells of each comparison. No N from 1 to 10 reverses this, although the gap narrows as N grows (Fig. 3).

Caching. Figure 5 shows why caching does not rescue aggregation. With its pairing and prefix caches, a hit in arm B still costs one pairing check, 578–589 μ s whatever N is in the medium profile, where a hit in arm D costs one Ed25519 verification, 24.5–33.8 μ s. The floors differ by the cost of the primitives: 531 μ s for one BLS verification against 20.6 μ s for one strict Ed25519 verification (Table 9). For the default instantiation at $N = 3$, medium, a verification takes 109 μ s without the prefix cache and 26.4 μ s with it.

Bytes. Figure 6 and Table 4 compare chain sizes. Aggregation saves bytes against the default only from $N = 2$ in the small, medium and large profiles, and from $N = 3$ when the chain carries an approval receipt. At $N = 1$ the aggregate chain is the larger one, by 13 bytes (61 with a

Table 3: Verdicts of the frozen plan: median latency ratio of the aggregating arm over the non-aggregating one, with its pooled bootstrap 95% confidence interval, and the verdict (–: not a net benefit). The same verdict holds for B/D in the prefix-miss state.

Profile	N	B/D, warm+prefix	A/C, warm	A/C-batch, warm
small	1	24.15 [24.15, 24.15] –	15.98 [15.97, 15.99] –	19.46 [19.45, 19.46] –
small	2	23.52 [23.48, 23.52] –	13.25 [13.25, 13.25] –	18.51 [18.51, 18.51] –
small	3	22.97 [22.97, 23.01] –	11.86 [11.86, 11.86] –	17.97 [17.97, 17.99] –
small	5	21.77 [21.77, 21.77] –	10.50 [10.50, 10.50] –	17.39 [17.39, 17.40] –
small	10	19.32 [19.32, 19.33] –	9.38 [9.38, 9.38] –	16.90 [16.89, 16.90] –
medium	1	23.61 [23.61, 23.61] –	14.68 [14.68, 14.68] –	17.40 [17.40, 17.40] –
medium	2	22.77 [22.77, 22.77] –	11.83 [11.83, 11.83] –	15.81 [15.81, 15.81] –
medium	3	22.00 [21.99, 22.00] –	10.52 [10.52, 10.53] –	14.91 [14.91, 14.92] –
medium	5	20.37 [20.36, 20.37] –	9.20 [9.20, 9.20] –	13.99 [13.99, 14.00] –
medium	10	17.40 [17.40, 17.41] –	8.25 [8.24, 8.25] –	13.27 [13.27, 13.28] –
large	1	20.47 [20.47, 20.50] –	7.60 [7.60, 7.61] –	8.27 [8.26, 8.27] –
large	2	18.21 [18.21, 18.21] –	5.81 [5.80, 5.81] –	6.59 [6.59, 6.60] –
large	3	16.56 [16.56, 16.56] –	5.18 [5.18, 5.18] –	5.95 [5.95, 5.95] –
large	5	14.49 [14.47, 14.49] –	4.86 [4.86, 4.86] –	5.81 [5.80, 5.81] –
large	10	12.20 [12.20, 12.20] –	5.18 [5.18, 5.18] –	6.71 [6.71, 6.72] –
medium-approval	3	23.01 [23.01, 23.01] –	12.99 [12.99, 12.99] –	17.14 [17.14, 17.14] –

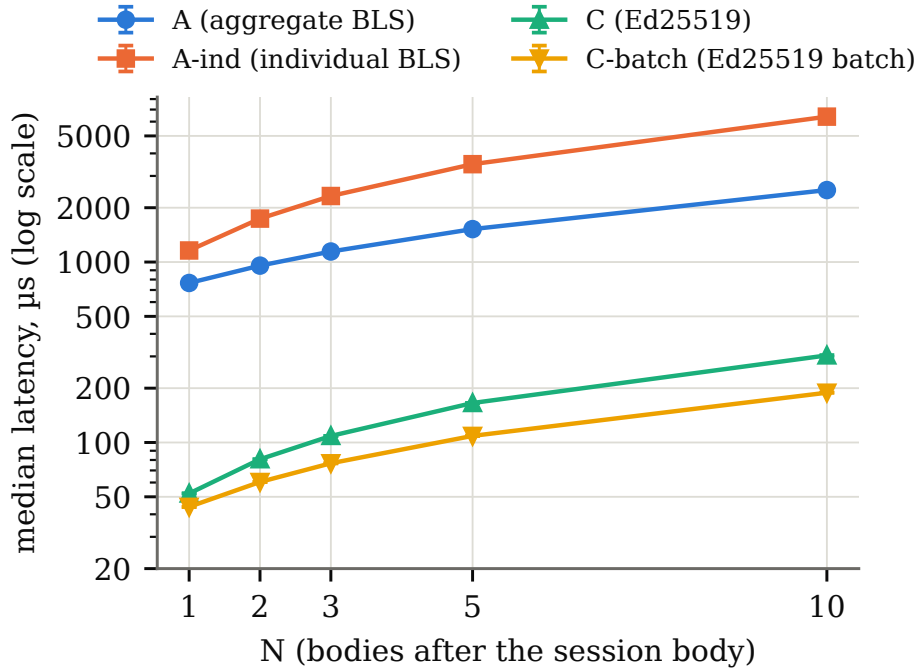


Figure 4: Warm median verification latency against N , medium profile, log scale: arms A (aggregate BLS), A-ind (individual BLS), C (Ed25519) and C-batch (Ed25519, batch verification). Points are pooled medians of three runs; error bars are the range of the three runs' medians.

receipt). The largest saving in the grid is 8.8% (small profile, $N = 10$); at $N = 3$ in the medium profile it is 2.9% (1826 against 1881 bytes). The saving stays small because the signatures are a small part of a chain: the 96-byte aggregate is 5.3% of arm A's chain at $N = 3$, medium, against 13.6% for C's $N+1$ signatures. Each body carries 48-byte BLS keys where Ed25519's are 32

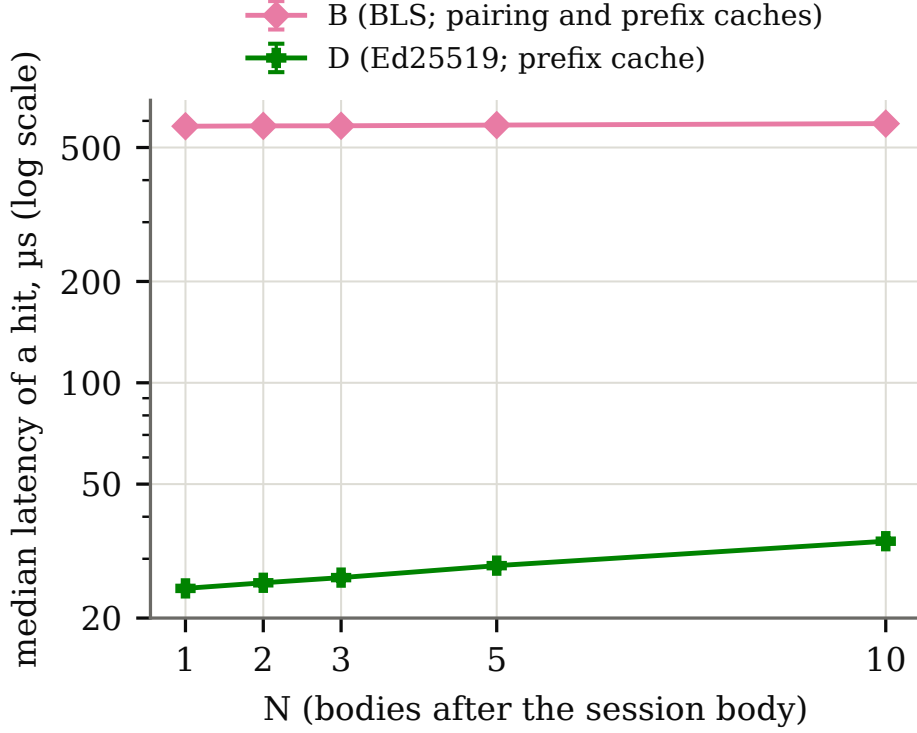


Figure 5: Median latency of a prefix-cache hit against N , medium profile, log scale: arm B (BLS with pairing and prefix caches) and arm D (Ed25519 with a prefix cache). Points are pooled medians of three runs; error bars are the range of the three runs’ medians.

Table 4: Chain bytes, arm A against arm C: the first N at which A’s chain is the smaller, the difference at $N = 1$, and A’s saving at $N = 10$.

Profile	Break-even N	A – C at $N = 1$ (bytes)	A smaller than C at $N = 10$
small	2	+13	8.8%
medium	2	+13	5.3%
large	2	+13	2.1%
medium-approval	3	+61	4.3%

bytes, and the bodies, with their policies, dominate.

Cost model. For arm A, warm, medium, the fit of Section 4.6 gives $\alpha = 566 \mu\text{s}$ and $\beta = 193 \mu\text{s}$, with $R^2 = 0.9999$. The per-hop term dominates beyond $N = \alpha/\beta \approx 2.9$, and β is 0.84–0.94 of one hash to $\mathbb{G}_2$ plus one Miller loop (224 μs) across the profiles, as expected when a multi-Miller loop shares work across pairs. Cold, β rises to 800 μs , because each hop also pays for its certificate.

Aggregation against individual BLS. Against individual BLS signatures (arm A-ind) aggregation is a net benefit in every cell: A/A-ind is 0.387–0.594 warm for $N \geq 2$ and 0.701–0.840 cold, and aggregation saves 98.0 bytes per hop. The multi-pairing does what it promises; it is the comparison with Ed25519 that it loses.

Cold path and network latency. At $N = 3$, medium, a cold verification makes 4 resolver calls and 1 policy-store call. With no injected latency it costs 3574 μs in arm A and 208 μs in arm C. With 1 ms per call the figures become 17,470 and 9456 μs , and with 80 ms per call

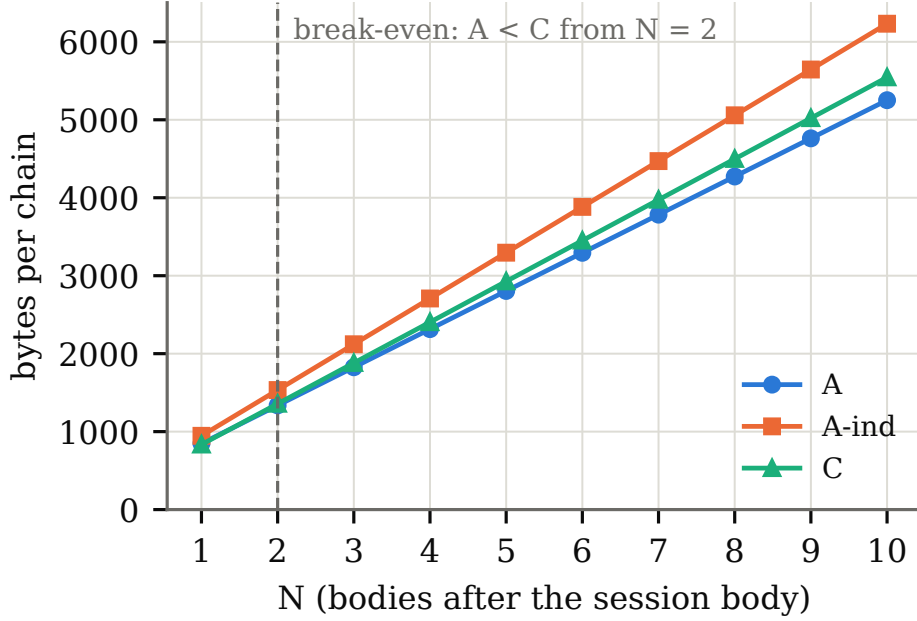


Figure 6: Chain size in bytes against N , medium profile: arms A, A-ind and C. The dashed line marks the break-even: A’s chain is smaller than C’s from $N = 2$. Sizes are means over 20 sampled chains and do not vary between runs, so there are no error bars.

469,530 and 445,618 μ s: network latency quickly dominates both. Verification is local only in the steady state (Section 9.2), and that steady state is what the verdicts measure.

Throughput. At $N = 3$, medium, arm D accepts 37,242 chains per second on one thread and 341,883 on ten, against 1712 and 15,924 for arm B; arm C accepts 9270 and 84,520, against 872 and 7997 for arm A. Adding the four efficiency cores (14 threads) raises the 99th-percentile latency per call from 1294 to 4092 μ s in arm A, and from 33.0 to 98.4 μ s in arm D. Table 9 in Section B gives the underlying primitive costs.

8.5 Design Claims Checked

Table 5 lists the design claims the plan undertook to check, with their verdicts. The claims and the criteria for judging them were fixed against the design as specified before measurement, when the plan was frozen. One claim was not supported: that cheap checks reject a hostile chain before any expensive cryptographic check. It holds for a warm verifier, but a cold one verifies $N + 1$ certificates first, and Section 4.6 now says so.

8.6 Positioning against Biscuit and AIP (exploratory)

Arm E maps DelegationChain’s chains onto Biscuit tokens (Eclipse Biscuit Project, 2026; Eclipse Biscuit Project, 2025). It lacks registry resolution, proof-of-possession, revocation, receipts, the nonce cache and parameter binding, so it is a positioning reference, not a competitor. The plan included one comparison involving it, against the figures AIP publishes for its chained mode (Prakash, 2026a), which also uses Biscuit. Measured here, arm E took 0.26–0.29 of AIP’s published time in the small profile, 0.32–0.37 in the medium profile and 1.11–1.26 in the large one. The two do not time the same operation. AIP’s code (Prakash, 2026b) shows that its timed verification decodes base64, verifies every block’s signature twice, parses its authorizer

Table 5: Design claims and their verdicts. The claims and the criteria for judging them were fixed when the plan was frozen, and the verdicts were judged against the text as it stood then. The “Paper” column gives the section of this paper where each claim now stands.

Claim	Paper	Verdict	Evidence
Aggregation gives significant constant-factor savings over verifying signatures independently	§2.2	Supported	A/A-ind is “net benefit” by the verdict rule of Section 8.3 in 16 of 16 warm cells and 16 of 16 cold cells. The ratio is 0.387–0.594 warm for $N \geq 2$, and the saving grows with N rather than staying constant.
Cost $\approx \alpha + \beta N$, with β one hash to G2 plus one Miller loop per hop; which term dominates is left to measurement	§4.6	Supported	R^2 0.9994–0.9999 warm. $\beta = 189\text{--}211\ \mu\text{s}$ is 0.84–0.94 of hash + Miller loop (224 μs). $10\beta/\alpha = 3.417$ [3.417, 3.418]: the per-hop term dominates by $N = 10$, from $N \approx 2.9$.
A chain carries a constant 96-byte signature regardless of N	§2.2, §4.8	Supported	A’s signature bytes are 96 at $N = 1$ and 96 at $N = 10$, in every profile.
Carrying certificates inline costs more per hop than aggregation saves	§9.2	Supported	A certificate is 249 bytes (BLS) in this encoding; aggregation saves 98.0 bytes per hop over A-ind. The 2026-09-29 text’s arithmetic (its §8.2) assumed 48 + 96 bytes per certificate, which understates it.
In the steady state, verification touches no registry	§9.2	Supported	Warm verifiers make no resolver or policy-store calls (<code>count-ops: tests/arms.rs</code> and <code>tests/security.rs::phase_ordering_count_ops</code>). Cold $N = 3$ verifications make 4 + 1 calls (Q5).
Cheap checks reject hostile chains before any pairing: warm verifier	§4.6, Figure 2	Supported	A warm verifier rejects line-34, expired and wrong-audience chains with no pairing (<code>tests/security.rs::phase_ordering_count_ops</code>).
Cheap checks reject hostile chains before any pairing: cold verifier	§4.6, Figure 2	Not supported	Line 24 verifies $N + 1$ certificates before phase 6, so a cold verifier pairs before rejecting a policy violation (P-28, D-61). Arm A costs 3573 μs cold against 1144 μs warm at $N = 3$.
The case for aggregation must be settled against a non-aggregating baseline	§8.3	Supported	The comparison changes the answer. Against individual BLS, aggregation wins (Q4); against Ed25519 it is not a net benefit in every cell: A/C 4.9–16.0 $\times$, B/D 12.2–24.2 $\times$.

from Datalog text on each call, and reports the mean of 100 single timings, each after creating a fresh token, on an older chip. That explains the direction of the gap and part of its size. To separate the rest, AIP’s own benchmark was run unmodified on this machine, outside the plan. By AIP’s own statistic, its code took 0.44–0.46 of AIP’s published times here. Both sides run the same code, so that factor cannot come from what is timed; it may come from the chip, the build, the toolchain or the operating system, which the run cannot tell apart. Against AIP’s code on this machine, arm E took 0.60–0.82 of its time (0.60–0.64 in the small profile, 0.75–0.82 in the medium one), within the factor of 3 that the plan set as a sanity bound for this comparison. AIP’s method is also sensitive to a cold start: in each build, the first process ran slower than the next two at the shallowest depths, and each of AIP’s published figures comes from a single such process.

Exploratory, and on the same machine, the default instantiation’s verification — which also checks parameter binding, receipts, revocation and replay — takes 109 μs at $N = 3$ in the medium profile, against 147 μs for Biscuit’s chained-token verification at matching depth (Table 6). Across N it takes 0.68–0.86 of arm E’s time, and with a prefix-cache hit 0.08–0.40.

8.7 Where Verification Time Goes (exploratory)

A separately instrumented build, never used for the results above, timed each phase of a warm verification at $N = 3$ for arms A and C in the small, medium and large profiles, over three runs on the same chains as the main measurement. The phases fall into five categories: decoding (line 2 and the canonical-form check, lines 4–6); policy (CONTAINS and EVALUATE, lines 32–37);

Table 6: Exploratory (not pre-registered). Medium profile, M9’s pooled medians on this machine, in μ s. DC’s N corresponds to Biscuit depth $N - 1$. Arm E (Biscuit) is a positioning reference, not a like-for-like arm: it has no registry resolution, no proof of possession, no revocation, no approval receipts, no nonce cache and no parameter binding (paper Table 1). The AIP column, also exploratory, is AIP’s own code (`bench_chained` at `ad2faa6`) measured on this machine in the exploratory session (D-79): the median of all its timings at the same Biscuit depth. It runs AIP’s own benchmark workload, not the medium profile; AIP’s benchmark stops at depth 5 (—).

N (Biscuit depth)	C, warm (μ s)	D, warm+prefix (μ s)	E (μ s)	AIP’s code here (μ s; exploratory)	$C \div E$	$D \div E$
1 (0)	52.2	24.5	60.8	81.4	0.86	0.40
2 (1)	80.8	25.5	104	133	0.78	0.24
3 (2)	109	26.4	147	183	0.74	0.18
5 (4)	166	28.6	232	282	0.71	0.12
10 (9)	304	33.8	444	—	0.68	0.08

identity (certificate lookup and checks, lines 23–28, without their cryptography); cryptography (point validation, the chain digests of lines 47–48, and signature verification at lines 24, 43 and 49); and everything else. We report each category’s share of the sum of the category medians. The shares agree across runs to within 0.7 percentage points; the absolute times vary more, and we do not report them.

Under the aggregate variant (arm A), cryptography is 88.1–99.1% of a warm call across the profiles, decoding 0.7–6.1% and policy 0.0–4.8%. Under the default instantiation (arm C), cryptography’s share falls as the policy grows, from 90.1% in the small profile to 39.1% in the large one, where decoding takes 30.9% and policy 25.2%. Identity is at most 0.2% in any configuration: with cached certificates, lines 23–28 are lookups. The calibration probe flagged 5 of the 18 configurations, and none was re-run; at that rate, the main measurement’s safety valve would have aborted the run.

8.8 Threats to Validity

The results come from one machine, an Apple M4 Max under macOS, with no core pinning or frequency control; measuring threads ran at the operating system’s user-interactive quality-of-service class, and throttling was watched by the operating system’s thermal report and by the calibration probe. `target-cpu=native` reaches the Rust code but not `blst`’s C and assembly, so the build flags help the Ed25519 arms more than the BLS ones; `blst` used its armv8 assembly, and the Ed25519 library its serial backend, so the ratios could differ on x86. The registry is in-process, and network latency was injected as sleeps, not measured over a network. The small and large policies and all workloads are synthetic; only the medium profile is the paper’s own example. C-batch’s batch verification has different semantics from strict verification at the edges. The multi-threaded supplementary arm warms the chip, so its own latencies include that thermal state; it appears in no headline ratio. The pooled confidence intervals are narrow because of pooling and understatement between-run variation, which the run-agreement rule guards against. Finally, arm E and AIP’s published figures measure different operations.

The measured binaries predate the canonical-encoding checks of Section 4.7. We measured the checks separately, in an exploratory run on the same machine. They are byte comparisons costing a few nanoseconds per signature or key: about 1.9 ns per signature and 1.2 ns per key on honest inputs, and at most about 11 ns on the worst inputs that pass. Across both arms of the default instantiation, every cache state, and $N \in \{1, 3, 10\}$, in the medium profile, they add under 0.1% to any measured median latency on honest inputs, and under 0.4% on worst-case inputs. The measured latencies therefore stand as the protocol’s cost.

9 Discussion

This section discusses related work, limitations of the protocol and its analysis, and directions for future work.

9.1 Related Work

DelegationChain draws on three distinct research traditions: capability-based authorization, pairing-based cryptographic protocols, and agent-system security. We position our contribution relative to each.

Capability-based authorization. The conceptual framework of attenuated capabilities, in which authority is represented by unforgeable tokens that can be progressively restricted as they are delegated, dates to early work on capability operating systems and was formalized in the modern context by Miller (2006). The most directly relevant prior work is Macaroons (Birgisson et al., 2014), which applied attenuated capabilities to cloud authorization using HMAC chains. A Macaroon carries “caveats,” predicates that restrict its use; each delegation appends additional caveats, and verification recomputes the HMAC chain using a shared secret known to the verifier. Macaroons demonstrated that attenuated delegation could be made cryptographic and efficient. Macaroons also provide *third-party caveats*, in which a caveat names a principal other than the target service and is satisfied by a discharge macaroon obtained from that principal (Section 2.3). This is the mechanism by which Macaroons reach across trust boundaries, and any comparison that treats Macaroons as a single-domain-only design is inaccurate. DelegationChain extends the model in three principal directions: from shared-secret HMAC to public-key signatures; from caveats that constrain future use to a binding of the specific invocation’s parameter values; and from informal caveat languages to a typed declarative policy language with termination and containment guarantees.

The substantive difference from third-party caveats is the shape of the verification path rather than its reach. Discharging a third-party caveat requires the holder to contact the named principal and collect a discharge macaroon before presenting the token; the third party must be available at that moment, and each additional third party adds a round trip. DelegationChain moves that authority into signatures that the verifier checks locally against registry-resolved public keys, so the request path carries no discharge-collection step. The cost of this choice is the identity infrastructure of Section 5, which Macaroons does not require.

UCAN (UCAN Working Group, 2026) also extends capability delegation to a public-key setting, originally for Web3 contexts. UCAN 1.0 signs tokens canonically encoded in DAG-CBOR, inside a Varsig-headed envelope, with Ed25519 recommended and P-256 and secp256k1 also required, and expresses delegation attenuation through scoped capability tokens. Our work differs from UCAN in three respects. UCAN 1.0 does specify a policy language over command arguments, but we did not find a defined containment relation between capabilities, which is what Section 6.4 supplies and what the scope monotonicity check requires. We target MCP rather than Web3 transport. And we include a federated identity infrastructure with proof-of-possession, policy ceilings pinned by the receiving organization, and revocation by identity–key binding. UCAN’s specification recommends memoizing validated delegation proofs, so that tokens sharing a proof need not recheck it (UCAN Working Group, 2026); that is the prefix caching the default instantiation of DelegationChain uses (Section 8.4).

Authorization certificates and delegation chains. The problem of composing delegation chains and computing the authority that survives composition is older than this literature. SPKI/SDSI (Ellison et al., 1999) defines authorization certificates in which the principal is a

public key rather than a name, and specifies a *5-tuple reduction* algorithm that composes a chain of delegation certificates into a single effective authorization by intersecting the authorization tags at each step, subject to a delegation bit controlling whether further delegation is permitted. Reduction is the direct ancestor of the scope-monotonicity check of Section 6.4, and the correspondence is close: SPKI intersects tags where we require containment between scope expressions. The differences are that SPKI computes the intersection, producing the resulting authority, where our verifier only checks a containment relation that the delegator has already asserted; that SPKI’s tag **-forms* (prefix, range, set) are a fixed and deliberately small constraint vocabulary, where our policy language is typed and admits conditions over named parameter paths.

X.509 proxy certificates (Tuecke et al., 2004) apply the same idea within Web PKI and were deployed at scale in grid computing. An end-entity certificate signs a short-lived proxy certificate, which may itself sign another, producing a delegation chain that a relying party validates as an ordinary X.509 path; attenuation is expressed through an embedded `ProxyCertInfo` policy field. This is the closest classical prior art to the certificate-chain-plus-policy design of Section 5. Two differences are worth naming. RFC 3820 leaves the content of the policy field opaque — it names a policy language by OID and carries its bytes, without specifying one — so nothing in the standard provides the decidable containment check our Section 6 is built around. And proxy certificate chains root in a CA hierarchy, where our registry model is federated and explicitly declines a global trust anchor (Section 5.1).

Biscuit (Eclipse Biscuit Project, 2026) is the closest contemporary system in the capability-token line. Biscuit tokens are public-key verifiable, support offline attenuation — a holder may append a block of additional checks at any time, restricting the token without contacting the issuer, and cannot remove a block without invalidating the signature — and carry an authorization language based on a purpose-built Datalog with specified evaluation semantics. Biscuit’s Datalog is more expressive than our policy language, and its formal treatment of evaluation is more developed than the sketch we give. The difference runs the other way on one axis: Biscuit does not bind a token to a specific invocation’s parameter values, since attenuation constrains future use rather than committing to one call. Section 8.6 positions a Biscuit mapping of our chains against the default instantiation. We regard Biscuit’s Datalog as the more serious policy design and note that a containment check over it would be a substantially harder problem than the one Proposition 2 addresses.

ZCAP-LD (Lemmer-Webber et al., 2026), developed in the W3C Credentials Community Group, expresses object capabilities as linked-data documents with invocation and delegation proofs, and shares the attenuated-delegation model. Its current draft attenuates in three fixed ways. A delegated capability may append a path or query suffix to its parent’s invocation target, must carry an expiry, and may allow only actions its parent allowed. That vocabulary is small and its containment is straightforward: a prefix check, a date comparison and a subset test. ZCAP is therefore not outside the decidability constraints of Section 6, as an earlier draft with extensible caveats would have been. What it lacks is any way to constrain an invocation’s arguments beyond a prefix of the target URL, with no range or membership conditions of the kind Section 6 provides. It also remains a community-group draft rather than a standard.

Proof-carrying authorization. A separate tradition places the burden of justifying access on the requester. Appel and Felten (1999) introduced proof-carrying authentication, in which a request arrives accompanied by a proof, in a higher-order logic, that it is authorized; the verifier need only check the proof, and the framework is expressive enough to encode Taos, SPKI, SDSI and X.509. Bauer (2003) developed the approach into a deployed system for web access control. The comparison is instructive because the trust structure is the same as ours — the verifier checks locally, without consulting the issuer — while the mechanism is the opposite:

proof-carrying authorization gains expressiveness by admitting an arbitrary logic and pushing proof search to the requester, where we give up expressiveness (Section 6.5) to keep the verifier’s work bounded by a syntactic check. A DelegationChain chain can be read as a very restricted proof, in which the only inference rules are signature verification and scope containment.

Cryptographic infrastructure. The BLS signature scheme is due to Boneh, Lynn, et al. (2001), and aggregation over distinct messages, with its security proof, to Boneh, Gentry, et al. (2003); that proof is the one the aggregate variant’s chain integrity argument uses. Rogue-key defenses for the same-message setting — proof-of-possession (Ristenpart and Yilek, 2007) and the alternative of Boneh, Drijvers, et al. (2018) — are not needed for it, and we use proof-of-possession only for attribution (Section 5.3). We use the signing variant standardized by the CFRG (Boneh, Bradley, et al., 2026), the BLS12-381 curve parameters of Sakemi et al. (2026), and the hash-to-curve construction of RFC 9380 (Faz-Hernández et al., 2023).

The problem our chain digest solves in the aggregate variant — producing a single compact signature over an *ordered* sequence of signers on distinct messages, such that the order is itself authenticated — is the problem that *sequential aggregate signatures* were introduced to solve. Lysyanskaya et al. (2004) gave the first construction, in which each signer folds its signature into the aggregate it receives from its predecessor, using certified trapdoor permutations; the order of signing is bound by the construction itself. Boldyreva et al. (2007) extended the line to ordered multisignatures and identity-based sequential aggregates in the pairing setting. We should therefore be precise about what is and is not new here. Ordered, aggregated signature chains are not new, and we do not claim them.

What we do claim is narrower: a specification of how, in the aggregate variant, to obtain order binding for delegation chains using a recursive hash chain over domain-separated token bodies together with *plain* BLS aggregation, rather than a dedicated sequential aggregate scheme. The reason for the choice is practical rather than theoretical. The construction needs no trapdoor permutations and no scheme-specific sequential signing operation; it uses the ordinary signing and aggregation operations of the CFRG BLS draft (Boneh, Bradley, et al., 2026), so it can be built on any conforming library, and the order binding lives in data the verifier already has to hash. What we give up is that the hash chain must be recomputed by the verifier from the received bodies, whereas a native sequential aggregate carries order binding inside the signature object. Section 8 settles the question of cost for our setting: our construction does not pay for itself against per-hop Ed25519 signatures, and with caching the comparison is decided by its verification floor of one pairing check per invocation. Any aggregate scheme whose verification needs at least one pairing check shares that floor; schemes built on trapdoor permutations (Lysyanskaya et al., 2004) do not, and we did not measure them. Certificate Transparency (Laurie et al., 2021) provided the model for our optional transparency log integration.

MCP and agent-system security. The security gap in cross-organizational MCP deployments has been empirically characterized by recent measurement work. H. Zhou et al. (2026) surveyed 7,973 live remote MCP servers and found 40.55% exposing tools without authentication; of the OAuth-enabled servers they were able to probe actively, every one exhibited at least one authentication flaw, and the disclosure process yielded nine CVE identifiers. Single-organization runtime enforcement approaches (Bühler et al., 2026) provide defense-in-depth within an organization’s perimeter but do not provide cross-organizational cryptographic proofs and are not designed to extend across organizational boundaries.

A second group of proposals secures the other direction of the MCP relationship. Maloyan and Namiot (2026) give a security analysis of the MCP specification and propose a backward-compatible extension — called MCPSec in their abstract and AttestMCP in the body of the paper — in which servers prove their claimed capabilities with certificates from a capability authority,

and JSON-RPC messages carry HMAC authentication bound to server identity. Metere (2026) similarly lets a host decide which tool servers, and which of their tools, it will admit. Both protect the host from untrustworthy servers. DelegationChain protects the server from unauthorized callers, so the two directions are complementary.

Closer to our direction, ACLE-MCP (Ding et al., 2026) addresses what its authors call the post-authorization execution trust gap: OAuth authorization does not ensure that a later call is executed by the workload the relying party meant to trust. For each protected call it issues a short-lived, sender-constrained capability lease binding the expected workload, the operation, object and parameter bounds, and downstream constraints; a provider-side gate consumes the lease before the tool runs. The differences are of mechanism and of scope. ACLE-MCP’s lease carries parameter bounds set when the lease is issued; our invocation token binds exact values, which the verifier then checks against the policy’s conditions. Its leases are issued per call, where our delegation prefix is built once and only the invocation token is signed per call, without contacting any issuer. Conversely, ACLE-MCP attests the executing workload, which DelegationChain does not address at all.

Llambí-Morillas and Fernández-Fernández (2026) formalize cryptographically verifiable agent authorization as a relation binding an agent principal, a concrete request, an execution context and the satisfaction of a policy, with a zero-knowledge proof of concept. Their request binding corresponds closely to what we call parameter binding. Their separation of request binding from runtime execution binding names the gap that ACLE-MCP addresses and that we leave open. Z. Zhou (2026) proposes governance requirements for agents that act across organizational boundaries — capability integrity, behavioral verifiability and interaction auditability. Among its results is that one unverifiable interior agent breaks end-to-end verification for every downstream node, the multi-agent counterpart of our requirement that every hop of a chain verify. Binding authorization to an invocation’s parameter values is not among its requirements.

The closest contemporary work is the Agent Identity Protocol of Prakash (2026a), which proposes invocation-bound capability tokens for delegation across MCP and A2A. AIP has two wire formats. Its compact mode, for single-hop cases, is an EdDSA-signed JWT carrying issuer, subject, scope, a budget, a maximum delegation depth and timestamps. Its chained mode is a Biscuit token with Datalog policies, in which each delegating agent appends a signed block that attenuates scope. Verification is local, with no authorization-server round trip. Identifiers either resolve over HTTPS (`aip:web`) or are the public key itself (`aip:key`), which is the same shape as our design. The authors list identity resolution, protocol bindings, completion blocks recording provenance, policy profiles and the compact mode as what AIP adds to Biscuit, and they ship interoperable Rust and Python implementations.

Three differences matter. First, despite the name, neither AIP format signs the parameter values of an invocation. Scopes name capabilities such as a tool, and — as with Biscuit, of which the chained mode is an instance — attenuation constrains future use rather than committing to a particular call. Parameter binding (Section 4.4) is the property on which the two designs differ most. Second, DelegationChain’s policy language has a formally specified containment check (Section 6.4); AIP’s Datalog policies, inherited from Biscuit, specify evaluation, and we did not find a containment check between them. Third, AIP has no registration step with proof-of-possession and no federation model, and it defers revocation to a later version, relying for now on short-lived tokens; Section 5 supplies registration with proof-of-possession, pinned policy ceilings, and revocation by identity–key binding. Both designs now sign each hop individually and have implementations; Section 8.6 positions them, with the caveat that the published figures and ours time different operations.

Workload and identity infrastructure. SPIFFE/SPIRE (SPIFFE Project, 2026) provides workload identity through short-lived X.509 certificates and JWT-format SVIDs. While SPIFFE

Table 7: DelegationChain against related systems. ✓ provided; p provided with qualifications (see notes); – not provided; u not specified by the cited source. “No anchor” abbreviates cross-organizational verification without a shared trust anchor.

System	No anchor	Param. bind	Aggregat.	Offline	Signature scheme	Attenuation	Policy formalization
DelegationChain	✓	✓	p	p	Ed25519 per hop; BLS12-381 aggregate variant	scope containment	typed grammar; termination proof; sound containment check
Macaroons (Birgisson et al., 2014)	p	✓	–	p	HMAC chain	caveat append	caveats, informal
UCAN (UCAN Working Group, 2026)	✓	✓	–	✓	EdDSA / ECDSA	capability attenuation	policy language specified; no containment relation
SPKI/SDSI (Ellison et al., 1999)	✓	p	–	✓	scheme-independent	5-tuple reduction	tag *-forms, specified
X.509 proxy certs (Tuecke et al., 2004)	–	u	–	p	X.509 (RSA / ECDSA)	embedded policy field	policy field opaque
Biscuit (Eclipse Biscuit Project, 2026)	✓	–	–	✓	Ed25519 / ECDSA-P256	offline block append	Datalog, specified semantics
SPIFFE (SPIFFE Project, 2026)	p	–	–	✓	X.509 / JWS	–	n/a (identity only)
OAuth 2.1 (Hardt et al., 2026)	–	–	–	p	JWS (bearer)	–	scopes, informal
AIP (Prakash, 2026a)	✓	–	–	p	EdDSA JWT / Biscuit	capability attenuation	Datalog, via Biscuit

addresses workload identity within a trust domain, it does not provide parameter-bound authorization or delegation chain integrity. OAuth 2.1 (Hardt et al., 2026) addresses caller identity and scope, may require synchronous token introspection on the verification path, and does not provide parameter binding.

For crossing organizational boundaries, the OAuth working group’s identity-chaining draft (Schwenkschuster et al., 2026) combines token exchange with the JWT authorization grant. The client’s authorization server in one domain issues a signed grant, which the authorization server in the next domain validates and exchanges for an access token, and the pattern repeats at each boundary. Scope can be narrowed at each exchange, so attenuated delegation across organizations is expressible in OAuth. The differences from DelegationChain are structural. Every boundary costs round trips to authorization servers on both sides. What reaches the resource is a token from its own domain’s authorization server, which vouches for the earlier hops rather than letting the resource verify them. And nothing binds the token to one invocation’s argument values. We view DelegationChain not as a replacement for these systems but as complementary infrastructure for the specific use case of verifiable agent-to-tool invocation across organizations.

Summary comparison. Table 7 collects the comparisons made above. The columns are the four properties this work set out to combine, plus the signature scheme, the attenuation mechanism, and the degree to which the policy component is formally specified. Aggregation turned out not to pay (Section 8), and DelegationChain keeps it only as a variant, which is why its own row marks that column p. We have marked a cell u wherever the cited source does not settle the question, rather than infer an answer from an implementation. The reader should note that our own row is not a column of ✓: offline verification is qualified for DelegationChain as it is for several others, for the reasons Section 9.2 gives.

Notes on the qualified cells. Macaroons is **p** on *no anchor* because the target service must hold its own root secret, while third-party caveats require no secret shared with the third party; it is **p** on *offline* for the same reason, since discharging a third-party caveat costs a round trip. SPKI/SDSI is **p** on *parameter binding* because authorization tags constrain values through **-*forms without binding a certificate to one invocation’s exact parameters. X.509 proxy certificates are **u** on parameter binding because RFC 3820 names a policy language by OID and carries its bytes without specifying one, and **p** on *offline* because path validation is local while revocation checking is not. Biscuit is **–** on parameter binding because attenuation constrains future use rather than committing to a single call. SPIFFE is **p** on *no anchor* because cross-domain use requires SPIFFE Federation. OAuth 2.1 is **p** on *offline* because a JWS access token verifies locally while introspection does not; the row describes the OAuth 2.1 core, and cross-domain chaining is discussed in Section 9.1. AIP is **–** on parameter binding because neither of its formats signs an invocation’s argument values, and **p** on *offline* because verification is local but **aip:web** identifiers resolve over HTTPS. DelegationChain is **p** on *offline* for the reasons given in Section 9.2, and **p** on *aggregation* because it is available as a variant (Section 4.8) that was measured not to pay (Section 8).

9.2 Limitations

We discuss limitations of the work along several dimensions.

Offline verification is qualified. We claim verification that requires no synchronous call to the issuing authority *on the dominant path*, and the qualification is load bearing. Two dependencies remain, and a third can be avoided. Certificate resolution (Section 5.4) queries the issuing registry on a cache miss. Revocation (Section 5.6) requires polling, push, or log monitoring, which is a continuous background dependency rather than a per-request one. Policy resolution is the avoidable one. Pinned policies can be fetched when they are pinned (Section 5.4), and only a verifier that defers the fetch to first use pays a round trip, on that call. In the steady state — a partner whose certificates and policies are already cached — verification touches none of these and is genuinely local. On a cold cache it is not, and statements elsewhere in this paper about offline verification should be read with that scope. A cold verifier also verifies $N + 1$ certificates before the policy phase (Section 4.6).

The obvious repair is to carry certificates in the chain, as X.509 does, so that a verifier needs the registry root key but not the registry. This would make verification offline modulo revocation, at a cost in bytes: in the reference implementation’s encoding a certificate is 185 bytes under Ed25519 (249 under the aggregate variant’s BLS keys), and a chain would carry $N + 1$ of them. We note it as a deployment option rather than adopting it.

Security arguments are sketches, not proofs. Section 7 presents proof sketches that reduce protocol security to standard cryptographic assumptions. These sketches have not been mechanized in a proof assistant nor have they been verified by independent cryptographer review. Symbolic protocol verification using Tamarin or ProVerif is planned and is expected to reveal at minimum the kind of subtle interaction issues that hand-analysis typically misses (e.g., key-compromise impersonation paths, unknown key-share variants). We treat the current analysis as a starting point for formal verification, not as a substitute for it.

Policy language expressiveness. The policy language of Section 6 is deliberately bounded. It cannot express semantic conditions requiring evaluation by an LLM or external system (“is this email professional?”), stateful conditions referring to data outside the invocation (“is this user under their monthly limit?”), or conditions requiring synchronous queries to external systems.

These limitations are intentional: each relaxation would compromise the decidability, termination, or side-effect-freedom guarantees on which the security analysis depends. Where richer policy expressions are required in practice, we anticipate them being enforced at the tool implementation layer *after* the protocol’s stateless check has succeeded. A fully-integrated treatment of stateful policy may require extensions to the protocol — including verifiable LLM attestation for semantic conditions — which we leave as future work.

Threat model boundaries. Several classes of threats are explicitly out of scope (Section 3.4), including fully-compromised HSMs, compromised signing service infrastructure, side-channel attacks, and attacks on human approvers’ devices. The protocol provides no defense against these and relies on operational measures (HSM hardening, key custody discipline, secure approver workflows) for mitigation. Within-policy prompt injection is bounded but not prevented (Section 7.3); the protocol’s contribution is to ensure that prompt-injected agents cannot exceed their permitted policy, not to prevent prompt injection itself.

Post-quantum security. Neither Ed25519 nor BLS12-381 is post-quantum secure. Per-hop signing lets the default instantiation adopt a post-quantum signature scheme such as ML-DSA (National Institute of Standards and Technology, 2024) without changing the protocol, at a cost in bytes per hop. The aggregate variant has no standardized post-quantum counterpart.

Evaluation. The evaluation’s own limitations — one machine, an in-process registry, synthetic workloads — are set out in Section 8.8.

Bootstrapping and adoption. The federated registry model requires both communicating organizations to operate compatible registries (or to participate in a shared transparency log). Adoption of the protocol in cross-organizational agent ecosystems thus faces a bootstrapping problem: early adopters have limited peers to communicate with, and the value of adoption grows with peer density. We do not address bootstrapping strategies in this paper, but observe that the same dynamic affects all cross-organizational identity protocols, including SPIFFE and SAML federation. Successful adoption is likely to require either integration into a widely-deployed platform (e.g., as a feature of an MCP-compliant agent framework) or coordination through standards bodies.

9.3 Future Work

We identify several directions for future work, in approximate order of priority.

Symbolic verification. Modeling the protocol in Tamarin or ProVerif and proving the security properties of Section 7 will validate (or, more usefully, falsify) the hand-written proof sketches, and can now also check the algorithm as implemented against the model.

Wider evaluation. The measurements of Section 8 come from one machine. Repeating them on Linux and x86, and in a networked deployment with a real registry, would test how far the verdicts carry.

Policy checks before identity resolution. The policy checks of lines 30–37 need no certificates. Running them before identity resolution would spare a cold verifier the certificate checks on policy-violating chains (Section 4.6), at the cost of renumbering the algorithm.

Containment completeness. Skipping child rules with unsatisfiable clauses in step 3(a) of the containment procedure would be sound, and would remove one of the sources of incompleteness that Remark 1 names.

A post-quantum instantiation. Instantiating Σ with a post-quantum scheme such as ML-DSA (National Institute of Standards and Technology, 2024), and measuring its cost in bytes and time per hop.

Differential privacy on audit trails. The protocol generates detailed audit records of every accepted invocation. For deployments in regulated industries, the ability to produce compliance attestations (“ $X\%$ of invocations were authorized”) without exposing the content of individual invocations would have practical value. Standard differential privacy mechanisms apply; their integration into the protocol’s audit format is straightforward but was deferred from this work for scope reasons.

Verifiable LLM attestation for semantic policy conditions. The most significant restriction of the policy language is its inability to express semantic conditions. Recent work on verifiable LLM inference (zkML and related approaches) may enable a future extension in which semantic policy conditions are evaluated by an LLM whose evaluation is itself cryptographically attested. This is speculative; the practicality of zkML at the latencies required for authorization protocols is not currently established.

Cross-registry naming and discovery. This paper does not address how organizations choose and namespace their `organization-id` values. In practice deployments will use DNS-style hierarchical naming or other conventions, but standardizing this is needed before adoption at scale.

10 Conclusion

Cross-organizational agent invocation needs authorization that a receiving organization can verify on its own: bound to the exact parameters of the call, constrained by policies both sides can reason about, and resistant to tampering with the chain of delegation behind it. DelegationChain provides this with digest-chained, per-hop signed delegation chains; a typed policy language whose containment check is polynomial and sound, with closed-world parameters and mandatory audience clauses; and a federated registry with proof-of-possession, pinned policy ceilings and revocation by identity-key binding.

We implemented the protocol, tested it against every threat in its model and against a differential oracle for its policy language, and measured it under a plan fixed before measurement. The measurement overturned the design choice the protocol was first built around: aggregating BLS signatures across the chain is not a net benefit in any configuration we measured, because a pairing-based verification floor costs far more than the bytes it saves. We report that result in full and specify DelegationChain over per-hop Ed25519 signatures instead.

The analysis remains a set of proof sketches, and the measurements come from one machine. Symbolic verification and a networked evaluation are the next steps. The design space for inter-agent authorization is still being mapped, and DelegationChain is one point in it, now a measured one.

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Use of AI Tools

The author used a large language model (Anthropic Claude) extensively in preparing this manuscript: to draft and organize text, formalize the policy language grammar, and prepare proof sketches. The model was also used as a critical reviewer across several revision passes, in which it identified technical and citation errors and proposed corrections, alongside review comments the author obtained from other sources. Some of the resulting corrections were adopted substantially as proposed, including the skip condition of the containment procedure (Section 6.4) and the policy audience clause (Section 6.3), and the model carried out the edits under the author’s direction. All choices among design alternatives, and the threat model, are the author’s. References were verified against original sources with the model’s assistance, including whether each supports the specific claim it is cited for.

The reference implementation, its test suites and the benchmark harness were written by a model-assisted coding agent (Anthropic Claude Code), working from a written specification and reporting to the author at fixed checkpoints. The specification was itself drafted with model assistance. The coding agent’s review of the specification before implementation, and the implementation itself, surfaced further defects in the protocol specification, among them a soundness gap in the containment procedure (found in review) and a gap in revocation (found in implementation), which this version corrects. The benchmark’s analysis plan was frozen before any measurement; every reported number is generated by the harness from archived raw data, and every deviation from the plan is recorded with the artifact. The design decisions, including the decision to respecify the protocol over per-hop Ed25519 signatures after the measurement, are the author’s. The author takes full responsibility for the content, accuracy, and conclusions of this work.

Data and Code Availability

The reference implementation, the specification, the frozen benchmark plan, the decision and deviation logs, and the generators for every table and figure in Section 8 are available at [\[PENDING: repository URL\]](#). The raw measurements are archived at [\[PENDING: Zenodo DOI\]](#); `ARTIFACT.md` in the repository describes how to verify the archives and regenerate every table and figure from them. Re-measuring under the frozen plan uses the measured commits that `ARTIFACT.md` names, not later builds.

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A Security Suite

Table 8 lists the security suite of Section 8.2, grouped by threat. “Asserted, in order” gives, for each test, the algorithm line whose rejection it asserts (Ln is line n of Algorithms 1 and 2), in the order the test asserts them, or acceptance where a threat is only bounded. The last two columns record the result of the test run that generated the table under the default instantiation and under the aggregate variant; — marks a test specific to the other instantiation.

Table 8: The security suite: each test, the algorithm lines it asserts, and whether it passed under each instantiation in the run that generated the table.

Test	Asserted, in order	Default	Aggregate
<i>T1a forgery</i>			
t1a_foreign_signatures	L49	yes	yes
t1a_victim_identity_signed_by_attacker	L49	yes	yes
t1a_victim_identity_with_attacker_key	L23	yes	yes
<i>T1b chain forgery</i>			
t1b_reorder	L11	yes	yes
t1b_truncation	L11	yes	yes
t1b_drop_session_or_invocation	L07 ($\times 2$)	yes	yes
t1b_extension	L07	yes	yes
t1b_insertion	L11	yes	yes
t1b_splice	L11	yes	yes
theorem_3_digest_recursion_catches_what_line_11_catches	phase 8 accepts the unmutated chain; L49 ($\times 3$)	yes	yes
<i>T2a / T2b</i>			
t2a_parameter_substitution_keeping_the_hash	L09	yes	yes
t2a_parameter_substitution_recomputing_the_hash	L49	yes	yes
t2b_delegation_wider_than_parent	L34	yes	yes
t2b_session_scope_exceeds_policy	L32	yes	yes
t2b_policy_not_pinned	L30	yes	yes
t2b_policy_unavailable_or_malformed	L31 ($\times 3$)	yes	yes
t2b_allow_all_under_a_rule_list	L34	yes	yes
<i>T3 reuse</i>			
t3a_outside_the_validity_window	L13 ($\times 2$); L15	yes	yes
t3b_replay_sequential	accept; L17	yes	yes
t3c_cross_context	L07 ($\times 3$)	yes	yes
t3d_cross_verifier	L08	yes	yes
t3d_audience_clause	L37	yes	yes
<i>T4 approvals</i>			
t4a_compromised_approver_is_bounded	accept (with approval); L37	yes	yes
t4a_approver_key_as_delegator	L26	yes	yes

Test	Asserted, in order	Default	Aggregate
t4b_routing_to_a_permissive_approver	L40	yes	yes
t4c_receipt_reuse	L43	yes	yes
t4d_stale_approval	L43	yes	yes
receipt_window	L44; L41; L42 ($\times 2$)	yes	yes
receipt_list_rules	L02 ($\times 3$); accept	yes	yes
<i>T5 identity infrastructure</i>			
t5a_misattribution_at_registration	registry rejects (BadPop); registry rejects (NonceUsed)	yes	yes
t5a_pop_under_the_chain_dst	registry rejects (BadPop)	—	yes
t5a_chain_signature_as_pop	registry rejects (BadPop)	yes	—
t5b_compromised_root_is_bounded	accept; L32; L24	yes	yes
t5c_revocation_propagation	accept; L27	yes	yes
t5c_expiry_and_not_yet_valid	L27 ($\times 2$); accept ($\times 2$); L27	yes	yes
t5d_certificate_substitution	L24; L25	yes	yes
t5e_role_confusion	L26 ($\times 3$); L41	yes	yes
scheduled_rotation_with_overlapping_certificates	accept ($\times 2$); L27 ($\times 2$); L42; L27	yes	yes
<i>T6 bounded misuse</i>			
t6a_prompt_injection_is_bounded	accept; L37	yes	yes
t6b_confused_deputy_is_bounded	accept; L34	yes	yes
<i>closed world</i>			
closed_world	L37 ($\times 3$); accept	yes	yes
<i>structure</i>			
structure	L02; L03; L05	yes	yes
structure_limits_and_classes	L02 ($\times 4$); L05	yes	yes
reencoding_alone_catches_non_canonical_bodies	L05 ($\times 2$)	yes	yes
point_validation_bls	L02 ($\times 2$)	—	yes
point_validation_ed25519	L02 ($\times 2$); L49; L02; registry rejects (InvalidKey)	yes	—
malformed_scopes_p15	L02 ($\times 2$); L31	yes	yes
<i>key chain, resolution</i>			
key_chain_keys	L18; L20	yes	yes
key_chain_identifiers_p16	L18; L20	yes	yes
resolution_unknown_signer	L23	yes	yes
distinct_messages	L48	yes	yes
<i>phase ordering, offline</i>			
phase_ordering_count_ops	L13; signature checks 0, pairings 0, resolver calls 0; L08; signature checks 0, pairings 0, resolver calls 0; accept; L34; signature checks 0, pairings 0, resolver calls 0; L34; signature checks 3, resolver calls 3; aggregate variant: hash-to-G2 3, Miller loops 6, final exps 3; accept ($\times 2$); default: signature checks 3, pairings 0; aggregate variant: hash-to-G2 3, Miller loops 4, final exps 1, signature checks 1	yes	yes
steady_state_is_offline	accept ($\times 2$); resolver calls 0, policy-store calls 0; cold: resolver and policy-store calls > 0	yes	yes

Test	Asserted, in order	Default	Aggregate
<code>theorem_5_part_2_cross_verifier</code>	accept; L08	yes	yes
<i>T3b replay, concurrent (Theorem 5)</i>			
<code>theorem_5_concurrent_replay</code>	accept; no outcome other than accept, L17, L50; exactly 1 accept per round; every other attempt L17 or L50	yes	yes

B Primitive Costs

Table 9: Primitive costs on the benchmark machine: criterion medians with 95% confidence intervals, in microseconds.

Operation	Median (μ s)	95% CI (μ s)
<i>BLS12-381 (blst)</i>		
verify (one signature)	531	[531, 532]
<code>aggregate_verify</code> , 2 messages	713	[712, 713]
<code>aggregate_verify</code> , 3 messages	894	[893, 894]
<code>aggregate_verify</code> , 4 messages	1075	[1074, 1075]
<code>aggregate_verify</code> , 5 messages	1257	[1256, 1257]
<code>aggregate_verify</code> , 6 messages	1438	[1436, 1438]
<code>aggregate_verify</code> , 7 messages	1617	[1616, 1619]
<code>aggregate_verify</code> , 8 messages	1799	[1796, 1801]
<code>aggregate_verify</code> , 9 messages	2034	[2032, 2034]
<code>aggregate_verify</code> , 10 messages	2219	[2219, 2220]
<code>aggregate_verify</code> , 11 messages	2405	[2404, 2405]
Miller loop	127	[127, 127]
final exponentiation	178	[178, 178]
hash-to-G2 proxy: context + <code>aggregate</code>	96.1	[96.1, 96.2]
pairing context alone	0.041	[0.041, 0.042]
hash to G2, estimated (proxy – context)	96.1	—
<i>Ed25519 (ed25519-dalek)</i>		
<code>verify_strict</code> (one signature)	20.6	[20.6, 20.6]
<code>verify_batch</code> , 2 signatures	32.0	[31.9, 32.0]
<code>verify_batch</code> , 3 signatures	39.7	[39.7, 39.7]
<code>verify_batch</code> , 4 signatures	48.2	[48.2, 48.2]
<code>verify_batch</code> , 5 signatures	56.6	[56.5, 56.6]
<code>verify_batch</code> , 6 signatures	64.4	[64.3, 64.4]
<code>verify_batch</code> , 7 signatures	72.8	[72.7, 72.8]
<code>verify_batch</code> , 8 signatures	81.7	[81.6, 81.8]
<code>verify_batch</code> , 9 signatures	89.5	[89.3, 89.5]
<code>verify_batch</code> , 10 signatures	98.0	[97.9, 98.1]
<code>verify_batch</code> , 11 signatures	106	[106, 107]